

Chapter 5

Post-CMP Cleaning

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5.1 CMP AND POST-CMP CLEANING OVERVIEW

Chemical mechanical planarization (CMP) combines chemical or electrochemical reactions with mechanical abrasion, in this case, polishing, to achieve material removal. The polishing process uses slurry to provide the abrasion action and various chemicals to provide the chemical reaction, dependent on the material to be polished. The slurry is applied to a rotating polishing pad, in which a wafer is in contact. This is the mechanism that polishes and planarizes the material on the surface of the wafer. Using CMP, the same material in different regions, such as silicon dioxide in field overburden areas and recessed trench areas, or different materials in the same region, such as silicon dioxide, silicon nitride, or polysilicon, can be selectively removed with the use of unique slurry chemical formulations, various polish downforces, and linear velocities of the polishing pad relative to the wafer. The effects of chemicals, CMP consumables, and kinetic factors on CMP material removal rate, *RR*, may be represented using Preston equation [1]:

$$RR = K_p PV \tag{5.1-1}$$

where *K_p* is the Preston coefficient, *P* is the polish pressure, and *V* is the relative velocity.

CMP was initially developed to improve planarity with respect to lithography depth-of-focus issue after interlevel dielectric (ILD) film deposition [2]. Nowadays, more and more CMP applications [3] are used for forming device structures (e.g., shallow trench isolation [STI], tungsten [W] local contact, copper [Cu] interconnect, aluminum [Al], or W replacement metal gate [RMG], FinFET [field effect transistor], and 3D NAND memory). Tight device structure control requires the CMP process to achieve near-perfect, within-wafer and within-die planarity across a surface that contains an array of geometric features with different pitch sizes and pattern densities. Fig. 5.1-1 shows a schematic of a typical configuration for a CMP tool polishing station. CMP performance optimization is a combination of consumable technology, such as slurry, polishing pad, conditioning disk, wafer retaining ring, and tool technology, such as slurry distribution, polish byproducts removal, velocity and downforce application, in situ removal profile adjustment, and polish endpoint control.

The chemical reactions occurring at the wafer–slurry interface may vary from simple hydrolysis reactions to more complicated chemical or redox

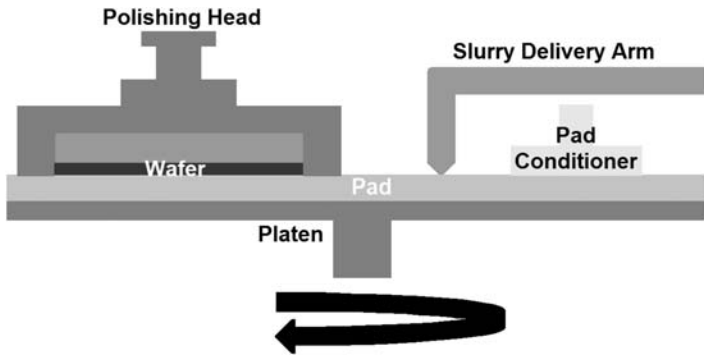


FIGURE 5.1-1 Schematic of CMP polishing station tool configuration. *Used with permission from the authors.*

reactions. The reaction products that build up on the field areas are removed by the friction between the abrasive slurry particles and the wafer surface, achieving successful planarization. However, the reaction products in recessed areas may act as inhibitor layers preventing surface oxidation or chemical etching required for successful polishing. These layers cannot be removed as a result of an insufficiently low mechanical force, which is desirable to prevent unwanted etching once planarization has been achieved. As a result, both selective film removal and planarization are achieved.

5.1.1 Post-CMP Cleaning Challenges

During CMP, the physical and chemical phenomena occurring at the wafer interface include reactant and product mass transfer, heat transfer (generated by chemical reaction and friction), momentum transfer (platen—head velocity effect), and chemical reactions. During polishing, abrasive particles and various chemical components are combined together onto the wafer surface, the ensuing chemical reaction products and mechanical wear debris (e.g., from pad and retaining ring wear) are generated. Additionally, removable defects (e.g., particles, organic residues, and metallic contaminants) and non-removable defects (e.g., scratches and corrosion) are generated on the wafer surface. CMP traditionally has been considered a “dirty” process in semiconductor manufacturing. Thus, post-CMP cleaning is required to displace all the removable contaminants, without generating additional defects or any other negative impact on device performance.

Post-CMP cleaning plays a critical role in meeting stringent CMP defectivity requirements. Fig. 5.1-2 shows examples of some defects observed after CMP. All defects, including abrasive particles, chemical additives, polish byproducts, pad/retaining ring debris, and metallic contaminants, must be removed from the wafer surface or prevented from occurring. The aggressiveness of

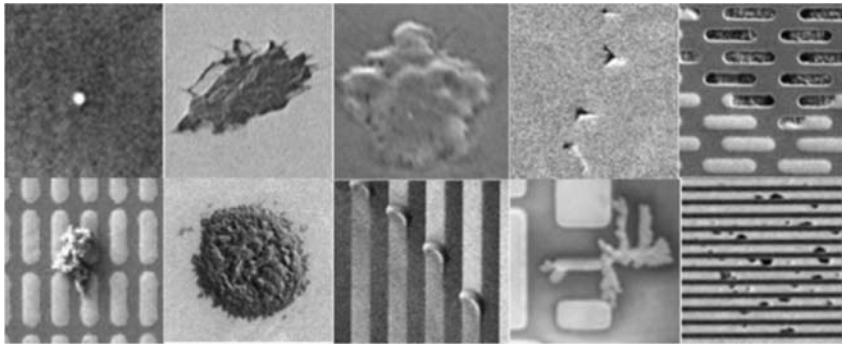


FIGURE 5.1-2 Examples of defects observed in CMP. Upper images from left to right: single slurry ball; fiber-texture organic residue; smearing polishing byproduct; scratches; missing metal defects. Lower images from left to right: agglomerated slurry particle; polishing organic byproduct; watermark on Cu lines; Cu dendrites; Cu corrosion. *Used with permission from the authors.*

mechanical forces and chemical reactions for cleaning depends on the mechanical and chemical property of all films present on the wafer surface. The adhesion forces between particles and the wafer surface is dependent on the pH and ionic concentration of the slurry, as described in Chapter 3. Furthermore, particle removal may become more difficult because of the delay of cleaning after wafer transfer from the polisher to the cleaner. Post-CMP cleaning must create clean and passivated surface. Cleaning itself should not generate watermarks, scratches, corrosion/galvanic corrosion, or surface roughness.

5.1.2 Post-CMP Cleaning Overview

Double-sided PVA (polyvinyl acetal $[C_4H_6O_2]_n$) roller brush scrubbing is widely used for post-CMP cleaning because of its high efficiency of removing loosely attached slurry particles. Megasonic cleaning has been added to improve particle removal efficiency, with the added benefit of organic residue removal because of megasonic power and temperature control. PVA disk scrubbing and jet spray cleaning are also used in some applications to supplement the cleaning capability of brush scrubbing. Spin rinse dryers (SRD) have been used for wafer drying. However, starting at the 90-nm technology node, the use of hydrophobic low-k dielectric (low dielectric constant) film applications have proliferated, thus isopropyl alcohol (IPA)-based drying technologies are now widely used. To improve particle removal efficiency, new cleaning approaches are being developed. Fig. 5.1-3 shows an example of a typical post-CMP cleaning sequence. The requirements for post-CMP cleaning are listed in Table 5.1-1.

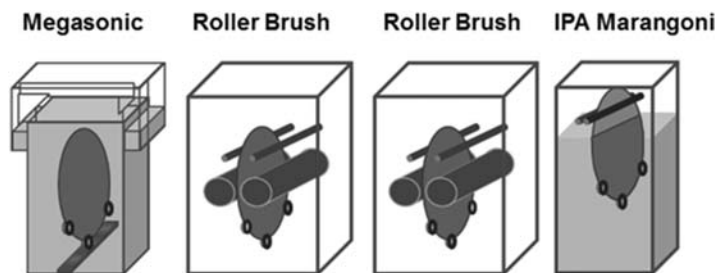


FIGURE 5.1-3 Example of a typical post-CMP cleaning sequence. From left to right: megasonic immersion module; brush box 1, brush box 2, and rinsing and drying module. *Used with permission from the authors.*

TABLE 5.1-1 Requirements for Post-CMP Cleaning

- Remove slurry abrasive particles (silica, ceria, and alumina particles)
- Remove particles generated during polishing
- Remove organic components of slurry chemical additives
- Remove metallic and ionic components from slurries
- Remove polishing byproducts generated by chemical reactions
- Remove organic residues generated by mechanical wear
- Remove metallic and reaction products from metal CMP
- Remove trace metallic contaminants
- Do not generate watermark and corrosion
- Minimize material loss and surface roughness generation
- Do not generate scratch and delamination
- Do not cause dielectric constant shift
- Obtain clean, stable, and passivated surface
- Achieve good electrical reliability (TDDB and electromigration) performance

5.1.3 Fundamental and Overview

5.1.3.1 Application of Cleaning Principle for Post-CMP Cleaning

Post-CMP cleaning has traditionally been based on the common etch/undercut/lift-off strategy to remove particles. The DLVO theory, named after Derjaguin, Landau, Verwey, and Overbeek, is applied to explain particle adhesion and particle removal by post-CMP cleaning technologies. Chapter 3 gives a thorough explanation of the theory. Typically, appropriate chemicals, sometimes with mechanical force, are applied to remove particles. These chemicals must not cause film damage. Particles that are adhered to the wafer through van der Waals force can be removed by causing electric repulsive forces to drive particles out of the electrostatic zone and to prevent particle reattachment onto the wafer surface. Modification or control of zeta potential has been widely used in the development of post-CMP cleaning solutions. Refer to

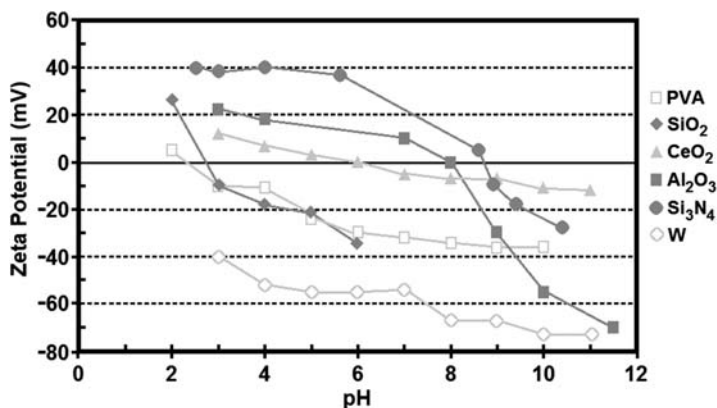


FIGURE 5.1-4 Zeta potential of common materials (PVA brush, SiO₂, CeO₂, Al₂O₃, Si₃N₄, and W) used in post-CMP cleaning. Data compiled by the authors from R. Vos, M. Lux, K. Xu, W. Fyen, C. Kenens, T. Conard, P. Mertens, M. Heyns, Z. Hatcher, M. Hoffman, *J. Electrochem. Soc.* 148 (12) (2001) G683; P. Supphantharida, K. Osseo-Asare, *J. Electrochem. Soc.* 151 (10) (2004) G658; I.J. Malik, J. Zhang, A.J. Jensen, J.J. Farber, W.C. Krusell, *Mat. Res. Soc. Symp. Proc.* 386 (1995) 109; J.G. Park, S.H. Le, H.G. Kim, H.D. Jeong, D.K. Moon, *Mat. Res. Soc. Symp. Proc.* 566 (2000) 173; K. Osseo-Asare, P. Supphantharida, in: R.L. Opila, C. Reidsema-Simpson, K.B. Sundaram, S. Seal (Eds.), *Chemical Mechanical Planarization IV, Electrochemical Society Proceedings*, vols. 2000–26, The Electrochemical Society, Pennington, New Jersey, 2000, p. 222. Used with permission from the authors.

Chapter 3 for details on zeta potential, van der Waals forces, and electrostatic forces. Fig. 5.1-4 shows the zeta potential of materials of interest to CMP cleaning, including brush, slurry particles, and thin film material [4–8]. The use of zeta potential plays an important role in particle removal and reattachment prevention. The three typical types of abrasive particles in CMP slurries are SiO₂ (silicon dioxide), Al₂O₃ (aluminum oxide), and CeO₂ (cerium oxide). The isoelectric point (IEP) of Al₂O₃ is at a pH value of about 8, which explains why it is more difficult to remove Al₂O₃ particles than SiO₂ particles, which have an IEP at a pH of about 3. In acidic solutions, ceria (Ce) particles are positively charged. Furthermore, Ce particles may also have chemical interactions with SiO₂ and polysilicon (polySi) surface making them more difficult to remove. Zeta potential values of the abrasive particles are used to explain the particle dispersion stability or agglomeration during polishing and cleaning. Through zeta potential control, electrostatic repulsion between the particle and the wafer is used to prevent particle reattachment during cleaning and rinsing.

For metal post-CMP cleaning, electrochemical potential–pH diagrams, known as Pourbaix diagrams [9], of the metal–water systems are used to explain corrosion that is observed during aqueous-based cleaning. The Pourbaix diagram shows the chemically stable species. Corrosion potentials for different

metals are used to guide the selection or development of cleaning solutions to minimize galvanic corrosion risk, which are discussed in Chapter 10.

As device geometry shrinks, there are more restrictions on the chemical aggressiveness of the solutions and on the mechanical forces that can be used with the different cleaning methods for post-CMP cleaning. New cleaning methods must have the capability to achieve high particle removal efficiency without sacrificing film stack integrity.

5.1.3.2 Device Nodes and Defect Inspection for CMP

Planarity, uniformity, and defectivity are the main attributes commonly used to judge CMP and post-CMP cleaning process performance. For qualification and process optimization of the CMP tool set and consumables, defects on blanket monitoring wafers can be monitored with light scattering-based defect detection instruments. As device feature size decreases, defects and particles with nanometer-size thresholds need to be measured. Defect inspection on patterned wafers can be performed with dark field or bright field-based metrology tools. Chapter 12 discusses the tools used for detection of defects and particle-based contamination. Wafer defect maps from defect inspection can be transferred to scanning electron microscopy (SEM)-based defect review instruments for defectivity classification and elemental identification using energy-dispersive X-ray spectroscopy (EDX). Defect review is important for troubleshooting root causes and providing directions for cleaning efficiency improvement. Defect inspection is typically performed immediately after CMP cleaning.

For sub-20-nm technology nodes, post-CMP cleaning is required to remove nanometer-sized slurry abrasive particles. These nano-sized particles may not be detected using the traditional defect inspection tools. Furthermore, the size of these defects may grow, as shown in Fig. 5.1-5, during subsequent film deposition steps, which is the so-called decoration effect. Nano-sized particles

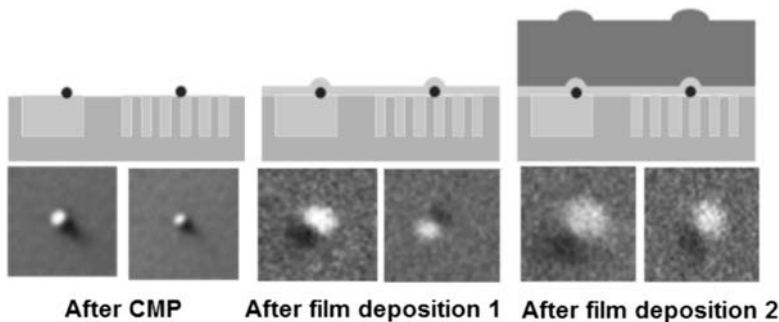


FIGURE 5.1-5 Schematic of nano-size slurry particles after Cu CMP and subsequent film growth showing the increase in size of the individual particles. *Used with permission from the authors.*

(individual slurry balls) also may cause shorts or opens in electric circuits, such as those shown in Fig. 5.1-6. Fig. 5.1-7 shows electrical data where the metal 2 layer shorts track linearly with the defect density [10]. For advanced technology nodes, post-CMP cleaning is required to reduce the numbers of total particles, the numbers of particles with smaller size thresholds, and defects that may not be detected immediately after CMP. Metallic contamination after CMP is measured by VPD ICP–MS (vapor-phase decomposition inductively coupled plasma–mass spectroscopy) or TXRF (total X-ray reflectance fluorescence) analysis. Chapter 13 outlines various analytical methods for surface analysis.

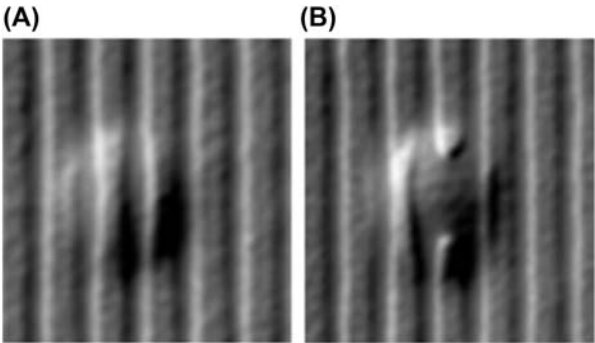


FIGURE 5.1-6 Example SEM images of defects caused by nano-sized slurry particles. Slurry particles left on the wafer may result in (A) shorts at the metal layer or (B) opens at the metal. *SEM images reprinted with permission from D. Jeanjean, O. Robin, R. Sramek, S. Mermoz, G. Ducotey, S. Gaillard, Y. Chen, F. Pitard, L. Nicoud, B. Brown, Proceedings of the 26th Annual SEMI Advanced Semiconductor Manufacturing Conference, 2015, p. 1. Copyright 2015, IEEE.*

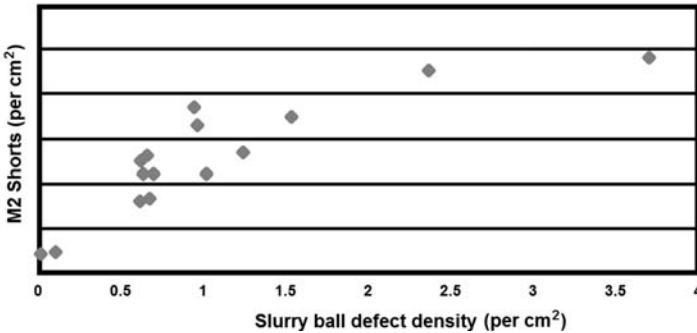


FIGURE 5.1-7 Correlation between electrical short at the metal 2 layer and nano-sized slurry particle density after Cu CMP [10]. *Reprinted with permission from D. Jeanjean, O. Robin, R. Sramek, S. Mermoz, G. Ducotey, S. Gaillard, Y. Chen, F. Pitard, L. Nicoud, B. Brown, Proceedings of the 26th Annual SEMI Advanced Semiconductor Manufacturing Conference, 2015, p. 1. Copyright 2015, IEEE.*

5.1.3.3 Removal of Slurry Particles and Polish Byproducts

Post-CMP cleaning is required to remove different types of slurry particles and polish byproducts, such as organic residues, as shown in Fig. 5.1-8, from different film surfaces. Silicon dioxide, Al_2O_3 , or CeO_2 abrasive particles, H_2O , and various chemical additives are commonly used to formulate CMP slurries. For metal CMP, polish byproducts may be complexes formed through chemical reactions of metallic compounds, corrosion inhibitors, and chelating agents. During the polishing process, the wafer surface is exposed to slurry and polish byproducts. Different polish byproducts may be generated with different slurries. Some byproducts may be chemically bonded to the film on the wafer surface. After polishing, both hydrophilic and hydrophobic films may be exposed. Immediately after polishing (during wafer dechucking from polishing pad, for example), slurry and polish byproducts are attached or pressed onto the wafer surface. Post-polish rinse with deionized water (DIW) may reduce the amount of slurry and byproducts, but a significant concentration of contaminants still remain on the wafer surface. The contaminants may be strongly adhered on wafer surface, as a result of incomplete water coverage of the wafer surface during the rinse or delayed cleaning during the transfer from the polisher to the cleaner.

5.1.3.4 Drying Challenges

Different film materials are encountered in CMP. After polishing, both hydrophilic and hydrophobic surface may be present on different device structures across the wafer surface [11]. Surface conversion from hydrophobic to hydrophilic through oxidation or surfactant adsorption may be allowed for

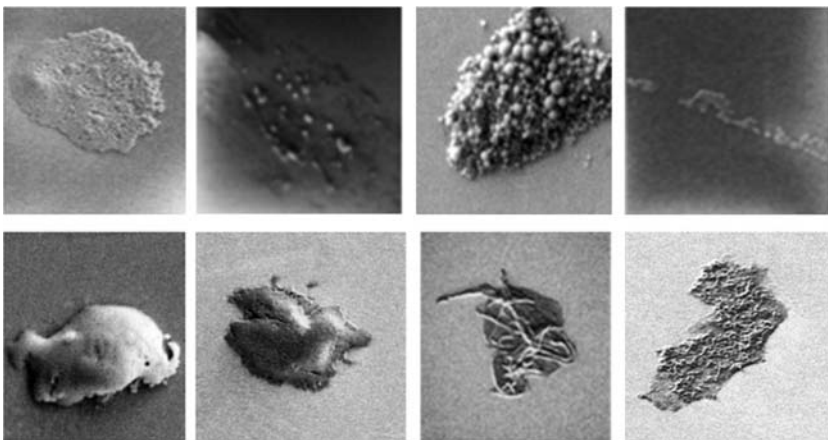


FIGURE 5.1-8 Examples of slurry particles (upper images) and polishing byproducts/organic residues (lower images). Slurry particles could be precipitated as agglomerated particles or attached to polishing byproducts/organic residues. *Used with permission from the authors.*

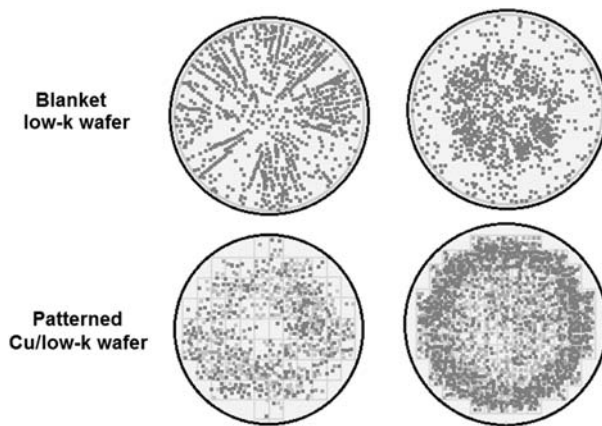


FIGURE 5.1-9 Defect scan maps after SRD drying of blanket hydrophobic wafer (upper maps) and patterned Cu/low-k wafers (lower maps). Different particle streak distribution signatures can be formed, depending on wafer surface hydrophobicity, Cu/low-k pattern structures and wafer rotation speed during SRD drying. *Used with permission from the authors.*

certain films, such as polySi [12,13]; however, for hydrophobic low-k films, surface conversion by chemical reaction or surfactant uptake may cause an increase in the dielectric constant. After cleaning, SRD had been widely used for drying hydrophilic wafers before the 90-nm technology node. On hydrophobic low-k films, water beads-up during wafer spinning and watermarks are formed. On a wafer surface with mixed hydrophilic and hydrophobic film combinations, such as Cu and low-k, because higher centrifugal forces are required to remove water layers from a hydrophilic surface and lower centrifugal forces are required to remove water layers from a hydrophobic surface (to avoid water beading-up), there is no process window for achieving watermark-free drying when using SRD. Fig. 5.1-9 shows defect scan maps after SRD drying of blanket hydrophobic and patterned Cu/low-k wafers [14]. Depending on device pattern structure, watermarks may be formed on hydrophobic low-k surface and also may be formed on hydrophilic Cu surface. Watermarks typically contain particles and residues on dielectric film surface. On Cu/low-k films, watermarks may also contain copper oxide particles, which cause significant degradation in device reliability. IPA Marangoni drying technology performed in a single-wafer configuration was developed and implemented to eliminate watermarks [15,16]. Fig. 5.1-10 shows Cu/low-k wafer defect performance comparison between IPA drying and SRD drying.

5.1.3.5 Defectivity and Passivation Challenges

Typically, defects generated by CMP may include film damage (e.g., scratches, line bending, or film rip-out), particles, and residues. The impact of these types of defect on device yield may be observed in the form of electrical circuit

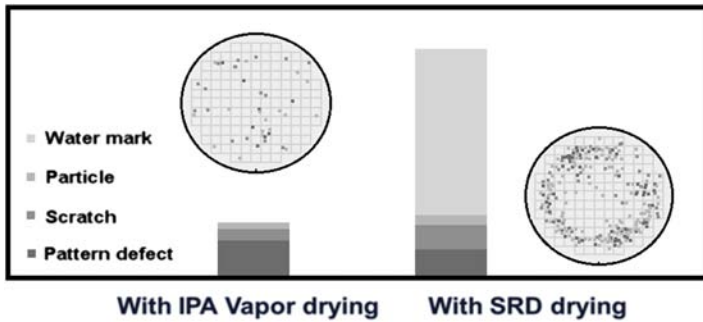


FIGURE 5.1-10 Copper/low-k wafer defect performance comparison between IPA drying and SRD drying. *Used with permission from the authors.*

shorts or opens. The impact of CMP and post-CMP cleaning on device electrical reliability typically is determined through time-dependent dielectric breakdown (TDDB) measurement. TDDB performance is an indirect indication of film surface quality and film stack interface integrity after CMP [17–23]. Factors associated with CMP that contribute to leakage are metal ion contamination on the surface, surface damage such as scratches, incomplete surface passivation such as dangling bonds, and high Cu surface roughness or Cu/barrier/dielectric side wall attack such as a slit divot. Fig. 5.1-11 shows a schematic of the impact of post-CMP wafer surface condition on TDDB performance. Dielectric film quality determines breakdown strength. Surface damage such as scratches may leave dangling chemical bonds on the dielectric film and may attract metal ions from liquid. These metallic contaminants left on film surface will cause dielectric breakdown at a low electric field.

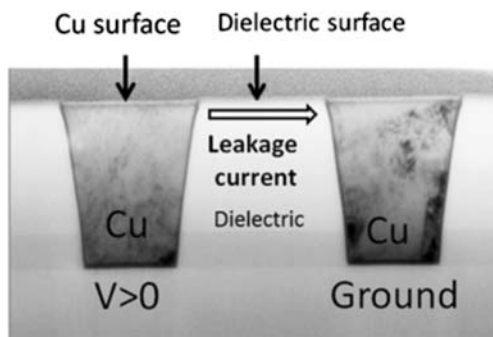


FIGURE 5.1-11 Schematic of the impact of post-CMP wafer surface condition on TDDB performance depicting the leakage that can occur between copper features. Dielectric film surface quality determines breakdown strength. $V > 0$ indicates positive voltage bias versus the grounded terminal. *Used with permission from the authors.*

For Cu CMP, post-CMP cleaning must generate a passivation layer on the Cu surface. Otherwise, Cu may be oxidized in air during the post-CMP queue time period before the next processing step, which typically is capping layer deposition. Although aggressive cleaning chemistries may demonstrate particle removal efficiency improvement, highly aggressive chemistries may remove the Cu passivation layer and cause reliability problem. Chapter 10 presents the passivation of Cu surfaces theory and fundamentals. On Cu/low-k patterned structures, watermarks generated during the SRD step would cause significant degradation in TDDDB performance [15], because of the presence of Cu ions. At the Cu/barrier-liner/dielectric film stack, such as the Cu/Co-TaN/low-k stack, the loss of Co would be observed as a slit divot at the Cu barrier–dielectric interface. The divot defects can be generated during polishing or cleaning because of either chemical etching or galvanic corrosion and may cause dielectric film breakdown or electric migration failure.

5.2 FEOL AND BEOL POST-CMP CLEANING

5.2.1 Dielectric Post-CMP Cleaning

Silicon oxide CMP includes polishing the STI films for the front-end-of-line (FEOL) and planarization of multilevel ILD films for isolating Al interconnect for the back-end-of-line (BEOL) integrated circuit manufacturing. High-pH slurries containing fumed or colloidal silica as abrasive particles commonly are used.

For STI, the polishing must be halted when the SiO_2 film is cleared and the silicon nitride (Si_3N_4) surface is exposed within the field area [24,25]. After CMP, the Si_3N_4 stop layer is stripped with a hot phosphoric acid solution, which may be considered a second cleaning step for post-polishing. Chapter 4 discusses aqueous cleaning methods.

For STI CMP or ILD CMP, brush scrubbing either with ammonium hydroxide (NH_4OH) or SC-1 (ammonium hydroxide [NH_4OH] and hydrogen peroxide [H_2O_2] mixture) solution can remove the silica abrasive particles from the oxide surface [26]. Adding megasonic cleaning before brush scrubbing for removing small particles and reducing brush particle loading (significant particle contamination within the brush material) improves the overall particle performance [27]. Hydrofluoric acid (HF), as a result of its SiO_2 etching capability, is used to reduce metallic contaminants, by stripping a thin, metal-contaminated SiO_2 layer off the wafer surface. Ideally, HF cleaning is performed after the particle removal with alkaline chemistry in the brush scrubbing step [28]. However, because of brush particle loading in the HF process and the relatively long DIW rinse time required after HF etching processing, HF cleaning is usually conducted before alkaline chemistry brush scrubbing. For advanced nodes, to achieve reduction of Si_3N_4 loss and minimization of feature topography, slurries with high removal rate selectivity

(SiO_2 vs. Si_3N_4) are needed. Therefore, slurries containing CeO_2 particles are the preferred oxide CMP slurries [5]. Neither brush scrubbing nor megasonic cleaning with dilute NH_4OH or SC-1 can effectively remove CeO_2 particles. HF cleaning in a brush box provides limited capability with respect to removing CeO_2 particles, which is achieved by stripping a thin CeO_2 -contaminated SiO_2 layer off the wafer surface. Wafer buffing with water or chemical on a polisher platen or a separate platen is required for more effective CeO_2 particle removal.

For high-k metal gate fabrication, the dummy gate-open CMP step includes polishing the SiO_2 film with highly selective slurries to stop on the Si_3N_4 surface [28]. This is followed by polishing of the oxide, nitride, and polySi films with a nonselective slurry to stop on polySi. For this process, the post-CMP cleaning step must remove particles and residues from the surfaces of oxide, nitride, and hydrophobic polySi film. Fig. 5.2-12 shows the sequence of (A) dummy gate-open CMP and (B) FinFET formation CMP. For FinFET dummy gate fabrication, the CMP step planarizes the polySi film with either ceria or silica-based slurries containing organic additives, which assist in the removal of the hydrophobic polySi film. Because the post-polish gate height stringently must be controlled, film material loss caused by post-CMP cleaning is not allowed.

For polySi post-CMP cleaning, highly diluted SC-1 solution with specific mixing ratios of NH_4OH versus H_2O_2 are used without generating significant surface roughness. As a result of the low cost associated with NH_4OH and H_2O_2 , oxide post-CMP cleaning has remained unchanged for a few technology node generations. However, for advanced nodes, such as the sub-20-nm

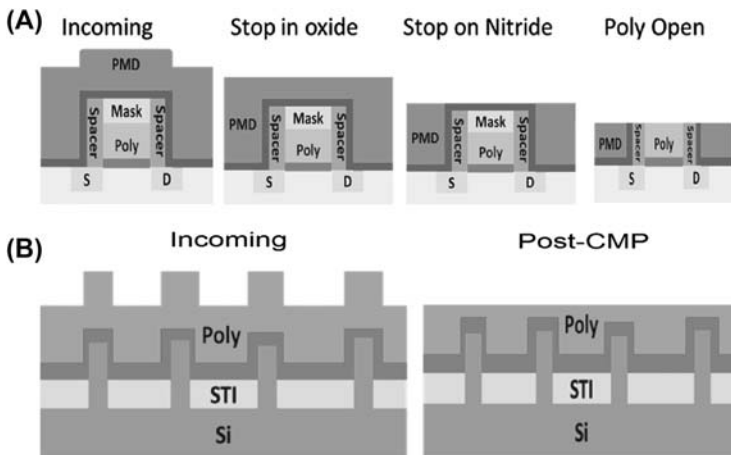


FIGURE 5.2-12 Sequence of (A) dummy gate-open dielectric CMP and (B) FinFET formation polySi CMP. D, drain of the transistor; PMD, pre-metal dielectric film; S, source. Used with permission from the authors.

technology, specialty acidic post-CMP cleaning solutions have been introduced for the FinFET polySi, etch stop layer, and ILD applications. In addition to metallic contaminant removal capability, the acidic cleaning solutions are formulated to improve wetting of hydrophobic polySi surface. However, brush scrubbing with acidic solutions is a challenge because of the requirement for minimizing reattachment of particles from brushes onto the wafer surface.

5.2.2 Metal Post-CMP Cleaning

Traditionally, processes similar to oxide post-CMP cleaning have been used for W contact or gate post-CMP cleaning [29,30]. The exception is that H_2O_2 cannot be used. Fig. 5.2-13 shows the W metal gate and metal contact polishing process flow. A short process time HF etching step followed by brush scrubbing with diluted NH_4OH is a common cleaning process. Megasonic cleaning with NH_4OH and low-concentration surfactants has been proven to be more effective than brush scrubbing for removal of organic residues inside alignment marks. Similar to oxide post-CMP cleaning, specialty acidic formulations are also used for W post-CMP cleaning. However, problems associated with brush particle loading or the loss of the surface passivation tungsten oxide film in an acidic environment have resulted in new-generation W post-CMP cleaning at near-neutral pH solutions.

For Al metal gate CMP, shown in Fig. 5.2-14, the Al film, the work function metals, and the barrier materials are polished away from the field oxide areas to isolate the individual transistor gates. The challenges of Al metal gate post-CMP cleaning includes the removal of residues containing Al_2O_3 , prevention of pitting corrosion on chemically active Al, and scratching the soft Al surface. Common acidic solutions for Cu post-CMP cleaning

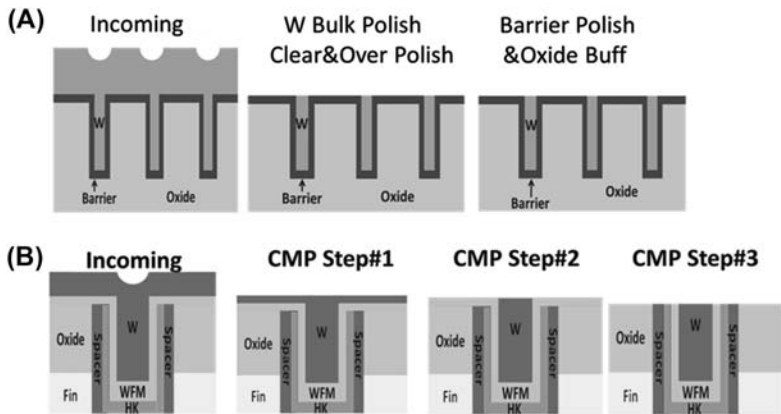


FIGURE 5.2-13 Sequence of (A) tungsten contact CMP and (B) tungsten gate CMP. HK, high-k material; WFM, work function metal material. *Used with permission from the authors.*

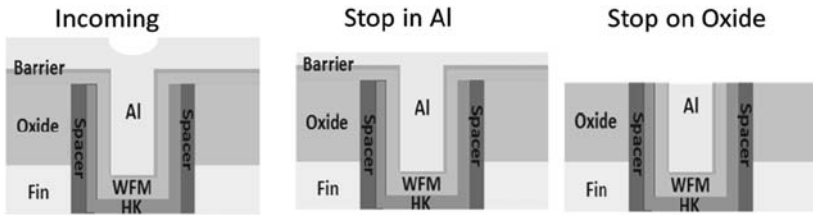


FIGURE 5.2-14 Sequence of aluminum metal gate CMP. *HK*, high-k material; *WFM*, work function metal material. *Used with permission from the authors.*

cannot effectively remove Al_2O_3 particles because of the positive zeta potential of Al_2O_3 at low pH. Alkaline Cu post-CMP cleaning solutions show enhanced capability for removing Al_2O_3 particles, compared with acidic solutions. However, Al etching or grain boundary highlighting exclude the use of alkaline solutions for Al gate post-CMP cleaning. Aluminum cleaning performance (e.g., particle removal, pitting, and scratching prevention) strongly depends on CMP slurries. Post-polish wafer rinse on a polish platen with certain chemicals may be used for Al cleaning.

5.2.3 Interconnect Post-CMP Cleaning

Copper CMP, shown in [Fig. 5.2-15](#), includes first a bulk Cu polishing step for overburden removal and planarization, then a Cu clearing step (Cu residue removal) for the field area, and finally a Cu barrier and dielectric film polishing step. Typically, the polishing is conducted sequentially with Cu slurry for the first two polishing steps and barrier slurry for the final step. If device wafers are processed through the post-CMP cleaner after polishing with Cu barrier slurries, then the cleaning process targets removal of particles and residues generated by the barrier CMP processing step. In many cases, for Cu polishing process qualification and monitoring, wafers immediately after Cu polishing also are sent to the post-CMP cleaner. Therefore, Cu post-CMP cleaning is required to provide cleaning capability after Cu polishing and after barrier and

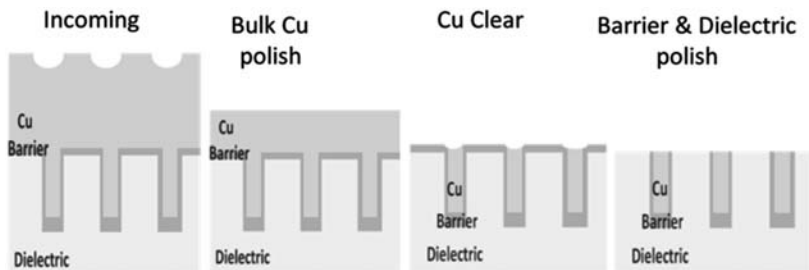


FIGURE 5.2-15 Cu CMP sequence. *Used with permission from the authors.*

dielectric polishing [31]. As a result of Cu CMP slurries containing various organic additives such as chelating agents and corrosion inhibitors, organic residue removal and corrosion prevention are the main challenges for these cleaning processes.

Scrubbing with soft PVA brushes has long been the dominant cleaning method for post-CMP. However, for advanced node Cu CMP, the brush compression required to achieve high particle removal efficiency, even with PVA brushes, may generate scratches on the Cu surface. Fig. 5.2-16 shows Cu surface circular scratches generated by brush scrubbing. Improvement in the PVA brush quality, such as geometry uniformity and cleanliness, and the Cu post-CMP cleaning solution is required to optimize cleaning performance and reduce scratches [32] created by brush scrubbing. It should be noted that brush circular scratches are more frequently observed on Cu and Al surfaces polished with slurries that do not generate protective oxide layers. During brush scrubbing, particles and residues on the wafer surface may be embedded into PVA brushes and generate scratches on the bare Cu or Al surface.

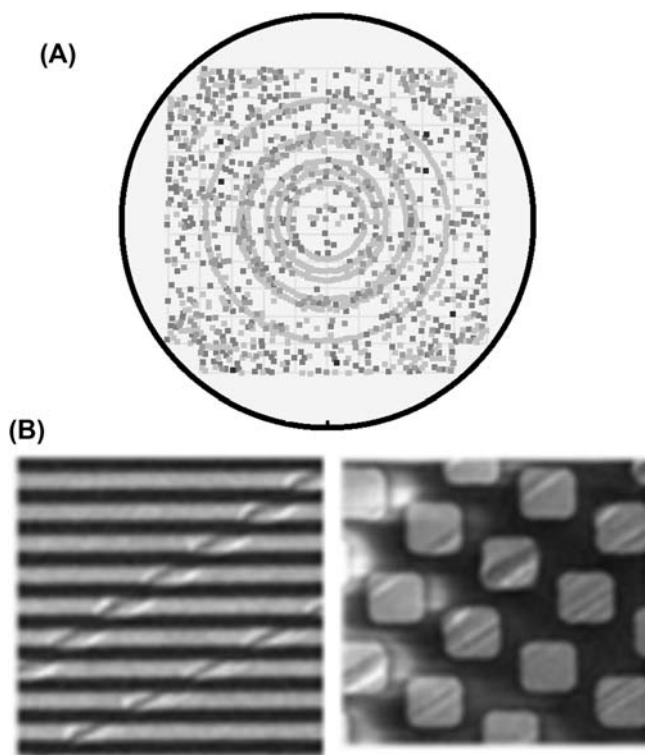


FIGURE 5.2-16 (A) Wafer scan map of copper circular scratches generated by brush scrubbing. (B) SEM images of the scratches. *Used with permission from the authors.*

Megasonic cleaning is used for Cu post-CMP cleaning. For challenging Cu barrier slurries, especially when a high concentration of benzotriazole (BTA) is used for Cu passivation, megasonic immersion cleaning with either acidic or alkaline chemistry shows better capability when compared with brush scrubbing for organic residue removal. Fig. 5.2-17 shows examples of (A) pattern

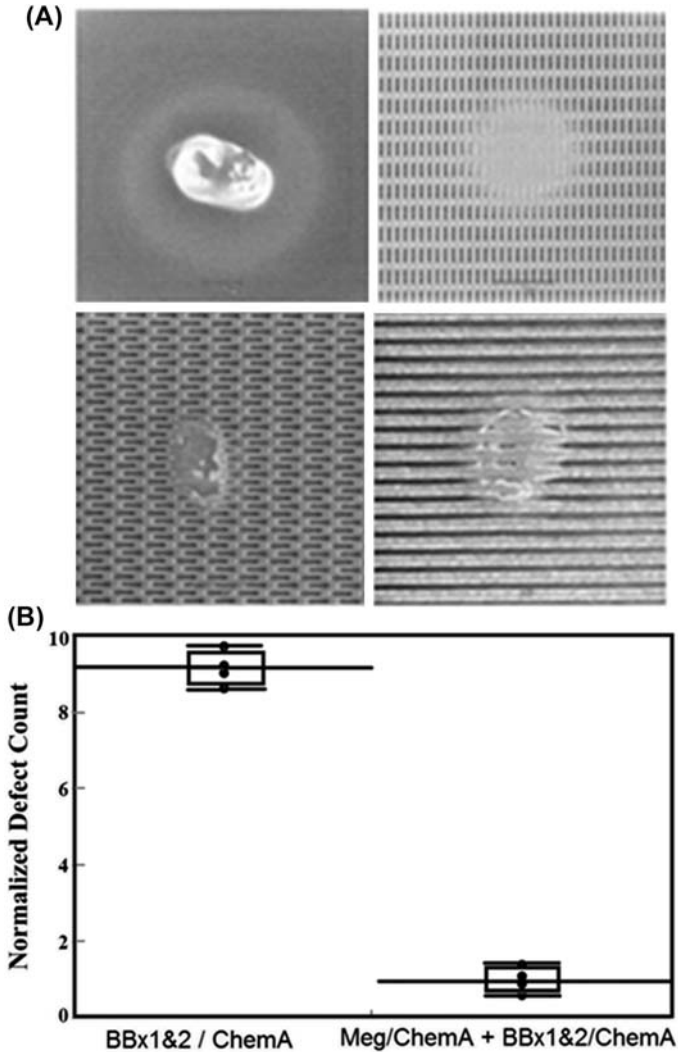


FIGURE 5.2-17 (A) Example SEM images of pattern defects generated by particles and residues from CMP. (B) A comparison of defects counts for Cu post-CMP cleaning with brush boxes only and the addition of megasonic cleaning for improving organic residue removal. *Chem A*, acidic cleaning solution; *BBx1&2*, brush box 1 and 2; *Meg*, megasonic immersion cleaner. *Used with permission from authors.*

defects generated by particles and residues added in CMP and (B) the cleaning results with the addition of a megasonic cleaning step before the brush cleaning step for improving organic residue removal.

The selection of acidic or alkaline chemistry for brush scrubbing depends on which Cu barrier slurries are used. Acidic cleaning solutions are selected for Cu post-CMP cleaning, for pre-20-nm nodes, mainly because of the good performance for removing organic residues and generating smooth Cu surfaces. Starting at the 20-nm node, alkaline chemistries are the dominant cleaning solutions as a result of improved Cu surface passivation. These chemistries form a thin layer of cuprous oxide on the Cu surface and are compatible with new Cu liner materials, such as cobalt (Co) and ruthenium (Ru). Since the introduction of low-k dielectric film for multilevel interconnect, IPA-based drying has been used for Cu post-CMP cleaning.

Prevention of Cu corrosion is important in Cu CMP and post-CMP cleaning. There are three types of corrosion in metal CMP or post-CMP cleaning: gross corrosion, galvanic corrosion, and photo-induced corrosion.

Gross corrosion is caused by chemical or electrochemical reactions between metal, such as Cu, W, Al, and Co, and the chemical components in slurries. Metal CMP slurries typically contain an oxidizer (H_2O_2) and corrosion inhibitors. During polishing, fresh slurry is dispensed onto the wafer–pad interface. Reaction products formed on protruding areas of the wafer are removed, while the passivation layers in recessed areas protect the metal. The dilution of fresh slurry by polishing byproducts or DIW may result in the reduction of corrosion inhibitor concentration; thus, corrosion may occur during polishing and post-polish wafer rinsing. If the metal surface is not well passivated, corrosion may also occur during wafer transfer from one polishing platen to another and during wafer transfer to the cleaning module and during post-CMP cleaning, especially when the oxidizer is not thoroughly rinsed from the metal surface. Chemical corrosion may be observed as gross corrosion, missing metal defects, voids, pits, dendrite growth, grain boundary decoration, or surface roughening. To prevent corrosion, slurry and cleaning solution composition optimization, platen-to-platen polishing time balance, post-polishing wafer rinse improvement, and wafer transfer or air-exposure time reduction are required.

Galvanic corrosion occurs when two dissimilar metals with different electrochemical potentials come in contact with each other within a film stack, such as Cu interconnect, seed, liner, and barrier materials, or if the metals are in different locations, they are connected through electrolyte solutions. In a film stack, because of galvanic corrosion, sidewall divot or fan shape recess defects may be observed. [Fig. 5.2-18](#) shows a schematic of a Co slit divot defect caused by Co/Cu galvanic corrosion. As a result of the thin Co layer, in this case 2- to 5-nm EDX elemental mapping of film stack cross section can be used to determine the depth of the slit divot. It should be noted for sub-10- μm nodes that Co is proposed to be used as a Cu barrier and seed enhancement

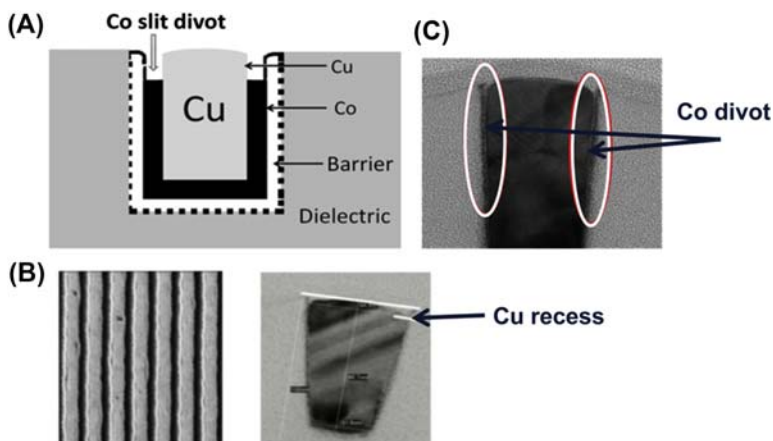


FIGURE 5.2-18 Examples of defects caused by cobalt liner attack by a copper barrier slurry. (A) Schematic of Co recess from Cu surface: slit divot defect. SEM and TEM image (B) and (C) of Co liner attack defect. Co slit divot may cause poor electromigration performance. *Used with permission from the authors.*

layer. Because of the large difference in the standard reduction potentials between Co and Cu (Cu^{2+}/Cu , 0.10 V vs. Co^{2+}/Co , -0.28 V), galvanic corrosion may occur when Co is in contact with the Cu during polishing and post-CMP cleaning.

Addition of corrosion inhibitors [33–35], such as BTA, is helpful in reducing galvanic corrosion. Optimization of the slurry or the cleaning solution formulation is required to minimize the difference in electrochemical potentials of the metals in a film stack, and thus to reduce galvanic corrosion risk. Fig. 5.2-19 shows examples of Tafel plots of Cu and Co for two copper barrier slurries. Minimizing the difference in Cu and Co corrosion potential by adding corrosion inhibitors can reduce galvanic corrosion risk. Film loss through galvanic corrosion also depends on the area of the galvanic-coupled metals. Because of the thickness of the Co liner can be only a few nanometers (<5 nm), electrochemical tests for mitigating galvanic corrosion must be conducted with film thickness close to that used in actual devices. As the thicknesses of metals in a film stack continue to decrease, it becomes more challenging to reduce galvanic corrosion [36–38].

Photo-induced corrosion only occurs on active device structures. This type of corrosion occurs because of the photons generated by light, which activates the movement of electrons and holes [39–42]. For example, photo-induced corrosion is observed on the Cu electrodes connected to the p region of a p-n junction, because of the built-in electrical field at the p-n junction. After the barrier layer is removed by CMP, the two electrodes are electrically separated from each other. The separation of the photo-generated electrons and

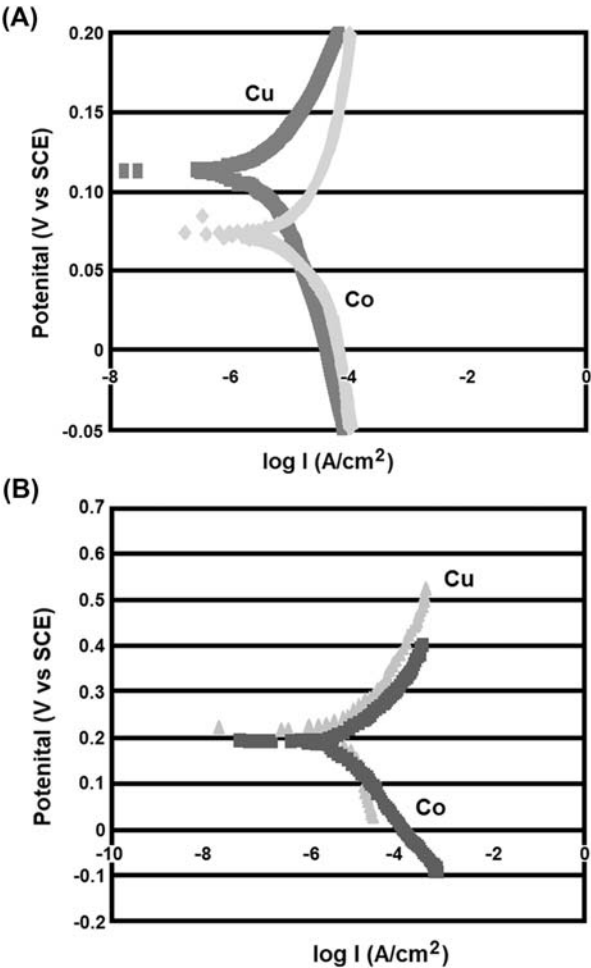


FIGURE 5.2-19 Examples of Tafel plots of Cu and Co in two copper barrier slurries. (A) The difference in Cu and Co corrosion potential may cause Cu/Co galvanic corrosion. (B) Difference in Cu and Co corrosion potential is minimized. *i*, current density; *SCE*, saturated calomel electrode. Used with permission from the authors.

holes by the electrical field generates electrical potential difference between the p and n electrodes. The positively biased Cu electrode corrodes, while Cu film deposition occurs on the negatively biased electrode, shown as a schematic in Fig. 5.2-20. Fig. 5.2-21 shows micrograph images of the p and n electrodes. Elimination of light exposure, formation of good passivation films, and reduction of electrolyte residues through improved wafer rinsing processes are required to prevent photo-induced corrosion.

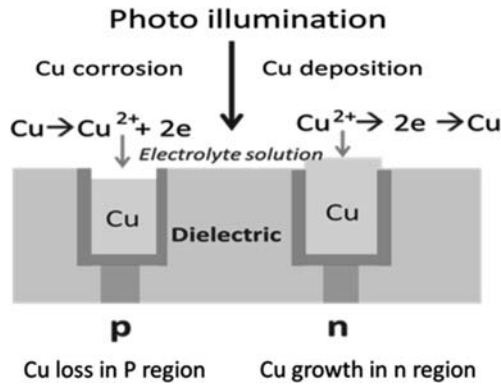


FIGURE 5.2-20 Schematic of photo-induced corrosion of Cu at the p and n electrodes, in this case the source and drain junctions. Cu oxidation happens in P region and Cu reduction happens in N region, in the presence of built-in device electric field, light exposure, and electrolytes, during post-CMP cleaning. *Used with permission from the authors.*

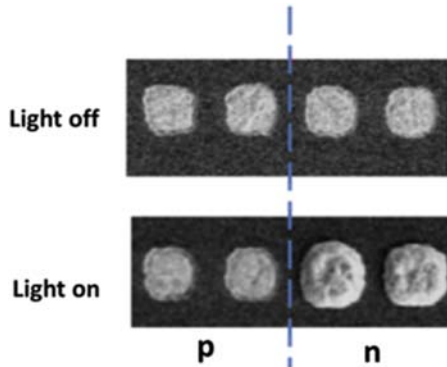


FIGURE 5.2-21 Micrographs of photo-induced corrosion of the p and n electrodes with and without exposure to light. Note the divot formed at the p electrode and the build-up of Cu at the n electrode when exposed to light. *Used with permission from the authors.*

5.3 POST-CMP CLEANING CHEMICALS

The role of post-CMP cleaning chemicals for particle and organic residue removal may include the following: dissolving organic residues and metallic contaminants, etching thin films to undercut the particles, creating electrostatic repulsion between particles and thin film, forming surface passivation layers, and improving hydrophobic surface wetting. The restrictions on post-CMP cleaning chemicals include the following: no corrosion, minimizing surface roughness increase, minimizing low-k dielectric constant shift, no removal of the surface passivation layer, no additional defects, such as brush circular

scratches, and no interaction between cleaning chemicals and polishing slurries or byproducts.

5.3.1 Diluted NH_4OH , TMAH, SC-1, and HF for FEOL Post-CMP Cleaning

Ammonium hydroxide, tetramethyl ammonium hydroxide (TMAH) $[(\text{CH}_3)_4\text{NOH}]$, and SC-1 are commonly used for the brush scrubbing and megasonic cleaning steps for FEOL SiO_2 post-CMP cleaning [43]. The oxide etching capability of these cleaning solutions assists with particle removal because of the undercutting effect required for particle lift-off. The high pH values of these solutions generate negative zeta potentials on most particles and thus create electrostatic repulsive forces between the particle and the substrate material. Ammonium hydroxide concentration from 1 wt% (weight percent) to 2 wt% are widely used for oxide post-CMP cleaning. However, highly diluted NH_4OH solution, ~ 0.05 wt%, also shows good cleaning capability. For SC-1 solutions, the concentration of H_2O_2 is typically $2\text{--}4\times$ of that of NH_4OH . Fig. 5.3-22 shows a schematic of particle removal mechanism [44–47] using brush scrubbing with alkaline solutions. Particles are removed through etching/under-cutting/lifting-off, dragging, and repelling. Brush torque values are used to quantify scrubbing mechanical strength, Torque:

$$\text{Torque} = \mu PAR \quad (5.3-2)$$

where μ is the friction coefficient, which is a function of rotation speed, chemistry, and wafer surface property, P is brush pressure, A is the area of contact, and R is the brush radius.

Cleaning with HF is based on its oxide etching capability to remove residual metallic and particulate contamination by lift-off. Fig. 5.3-23 shows a

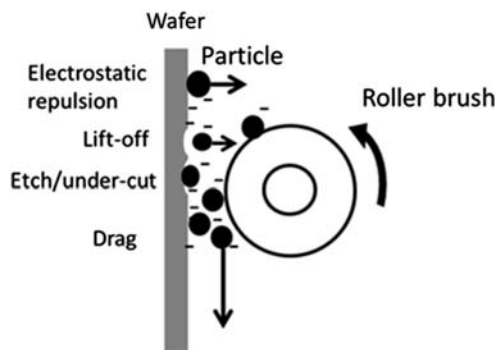


FIGURE 5.3-22 Particle removal mechanism of brush scrubbing with alkaline solutions. Particles are removed through etching/undercutting/lifting-off, dragging, and repelling. *Used with permission from the authors.*

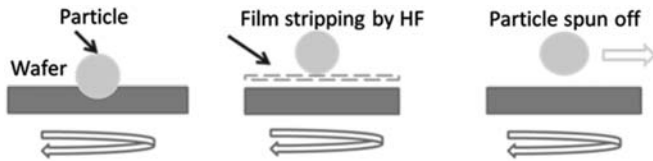


FIGURE 5.3-23 Schematic of particle removal by HF chemical only. *Used with permission from the authors.*

schematic of the lift-off mechanism for removing particles with only HF chemical treatment. The typical mixing ratios of HF to DIW used in post-CMP cleaning are 1:50 to 1:100. The high SiO_2 etching rate with HF results in the stripping of the top SiO_2 layers, which not only results in the removal of particles on the oxide surface but also results in the removal of metallic contaminants, however at the expense of removing some of the oxide film. The enlargement of microscratches because of oxide etching by HF is called the defect decoration effect. However, if oxide loss by HF etch is minimized, then the combination of HF undercutting etching and brush scrubbing lift-off can provide superior cleaning results, which cannot be achieved only by brush scrubbing with chemicals, such as NH_4OH or SC-1. Fig. 5.3-24 shows a schematic of the lift-off mechanism associated with HF chemical treatment along with brush scrubbing.

Cerium oxide particles have a high affinity to SiO_2 , Si_3N_4 , and polySi surfaces and are difficult to remove by brush scrubbing or megasonic cleaning with NH_4OH or SC-1 solution. Ce^{3+} is stable and soluble in acidic solutions; thus, cleaning approaches based on chemical reactions for Ce particle removal have been investigated [48]. However, using mild chemicals for Ce particle removal has not yet been demonstrated with wafer-level particle performance data. One alternative approach to removing Ce particles is to perform additional cleaning in wet benches with a sulfuric acid (H_2SO_4) and H_2O_2 mixture (SPM), also called Piranha solution. This solution is commonly used for photoresist

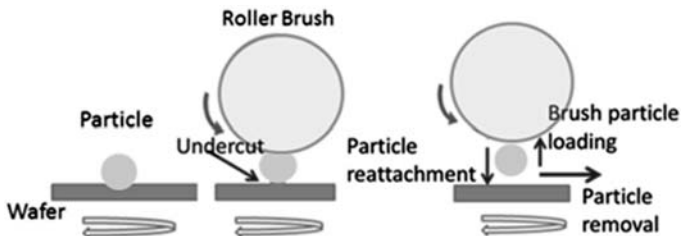
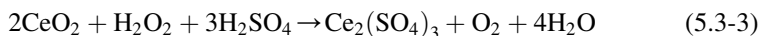


FIGURE 5.3-24 Schematic of the lift-off mechanism of brush scrubbing for removing particles with HF chemical treatment. *Used with permission from the authors.*

stripping at high temperatures ($>120^{\circ}\text{C}$), as discussed in Chapter 4. In SPM solutions, Ce^{4+} is reduced to Ce^{3+} by H_2O_2 . The Ce^{3+} is dissolved by H_2SO_4 :



The high reactivity of SPM and the high temperature required for reducing and dissolving cerium oxide make it difficult, from cost consideration, to implement this Ce particle removal process into high-volume production within an integrated post-CMP cleaning module. If sufficient film removal is allowed, through etching SiO_2 layers, cleaning with HF provides sufficient capability for removing Ce slurry particles, with or without brush scrubbing. Although brush scrubbing provides the mechanical force required for lifting off particles after HF undercut [5], particle loading onto brushes and particle redeposition back onto the wafer surface during HF etching and DIW rinse step are typically observed. For cleaning with HF, the required oxide thickness removed using 0.5–1.0 wt% HF can be achieved within acceptable cleaning time, for example, 30 Å oxide removal with a 30-s HF etch process.

For polySi post-CMP cleaning, HF is used for metallic contaminant removal, by etching the metal-contaminated oxide layer, but is not widely used for slurry particle removal because HF alone does not etch polySi film. Because brush scrubbing with NH_4OH or SC-1 cannot remove Ce particles effectively, a process using HF to etch native or chemical oxide layer, which are formed after polishing to undercut particles, combined with brush scrubbing can provide limited Ce particle removal capability. The challenge is to prevent particle reattachment onto the hydrophobic polySi surface after HF etching.

5.3.2 Acidic and Non-SC-1 Alkaline Solutions for FEOL Post-CMP Cleaning

For advanced nodes, such as the sub-20-nm technology, to increase the compatibility of cleaning solutions with CMP slurries and to take advantage of the cleaning capability of acidic solutions in organic residue and metallic contaminant removal, specialty organic acid-based acidic cleaning solutions are used for FEOL post-CMP cleaning, especially for polySi polishing processes. The acidic cleaning solutions are also formulated to improve wetting of hydrophobic polySi surface. However, there is a significant challenge when using acidic cleaning solutions because of the electrostatic attraction force between particles and films. Also, acidic chemistries have long brush break-in time and high brush particle loading in high-volume manufacturing. Fig. 5.3-25 shows a comparison of brush break-in with an acidic solution and an alkaline solution (SC-1). An alkaline solution can more effectively clean brushes and reduce the number of particle adders from brushes onto the silicon wafer surface than an acidic solution.

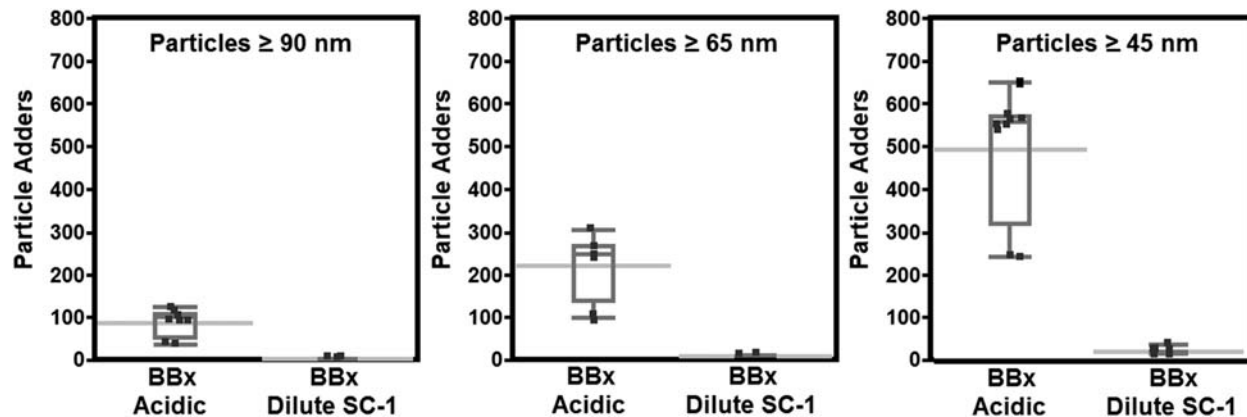


FIGURE 5.3-25 A comparison between an acidic solution and an alkaline solution (SC-1) used for brush break-in. Adders of three different size particles on Si wafers after 50-wafer brush break-in with a common acidic cleaning solution and diluted SC-1 solution. BBx, brush box. *Used with permission from the authors.*

To take advantage of the favorable zeta potential in the high-pH region, non-SC-1 alkaline solutions with metal-complexing agents show promise for reducing particle adders through particle reattachment prevention and for achieving metallic contaminant removal.

5.3.3 Ammonium Hydroxide and Neutral Solutions for Tungsten Post-CMP Cleaning

Ammonium hydroxide is widely used for W post-CMP cleaning in megasonic module and brush boxes [49]. In the absence of an oxidizer, W corrosion risk is minimized for W CMP when used for local contact formation. However, for sub-10-nm nodes, W CMP is widely used for creating W RMG, where W CMP defectivity requirements become more stringent than previous device nodes. Tungsten is highly susceptible to corrosion during post-CMP cleaning [50] because of the presence of H_2O_2 that is carried over from the polishing step. From the Pourbaix diagram of the W–H $_2\text{O}$ system [9,51,52], shown in Fig. 5.3-26, in the high-pH region, tungsten oxide dissolves and the unoxidized W surface is exposed; accordingly, W corrosion or W plug pinhole decoration may occur in alkaline solutions. Although alkaline solutions are good for particle removal from the SiO_2 surface, corrosion passivation agents are needed for reducing corrosion defects. In the low-pH range, tungsten oxide (WO_3) is stable and protects the underlying W film. Therefore, W is more

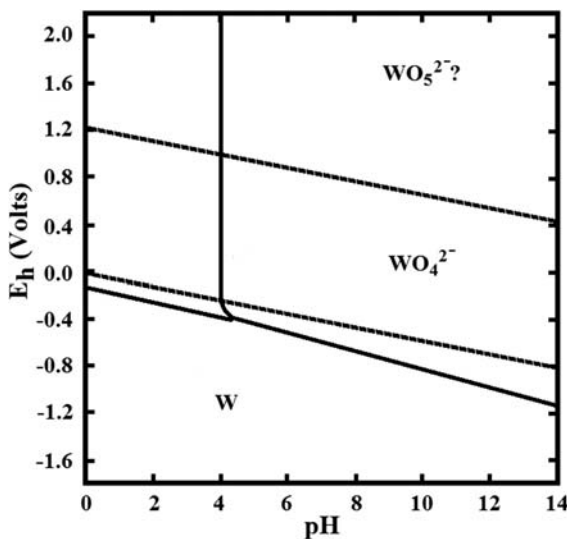


FIGURE 5.3-26 Potential–pH diagram of the tungsten–water system. Adapted from M. Pourbaix, *Atlas of Electrochemical Equilibria in Aqueous Solutions*, National Association of Corrosion Engineers, Houston, Texas, 1974; F. Zhang, A.A. Busnaina, G. Ahmadi, *J. Electrochem. Soc.* 146 (7) (1999) 2665. Reproduced by permission of National Association of Corrosion Engineers.

immune to corrosion in acidic chemistry. Acidic cleaning solutions, such as citric acid ($\text{C}_6\text{H}_8\text{O}_7$)-based solutions, can be used for W cleaning. However, low particle removal efficiency or particle loading issues are observed with these solutions. To reduce particle loading, cleaning solutions with neutral pH and additives, such as complexing agents and surfactants for the modification of Al_2O_3 abrasive particle zeta potential, are used for W post-CMP cleaning. To avoid W corrosion and achieve high particle removal efficiency, W post-CMP in neutral pH environment is being investigated.

5.3.4 Aluminum Metal Gate Post-CMP Cleaning Solutions

Aluminum polishing byproducts have high affinity to the Al surface and are difficult to remove [53]. Also, after polishing, if an adequate surface passivation layer is not formed on the Al surface, brush circular scratches can be generated. Post-polishing on-platen chemical buff (i.e., wafer chemical rinse on a polish pad) thus is used for Al CMP; such as Al surface passivation or particle/organic residue removal.

Fig. 5.3-27 shows a simplified Pourbaix diagram of the Al–H₂O system [9]. In low-pH solutions ($\text{pH} < 4$), the Al_2O_3 protective layer may be dissolved. Post-polishing chemical buff with acidic solutions in the low-pH region may cause Al corrosion and accentuate brush scrubbing ring scratches. In the pH range from 4 to 9, dense Al_2O_3 film on the surface protects the bulk Al underneath. Aluminum CMP slurries contain alumina (Al_2O_3) abrasive

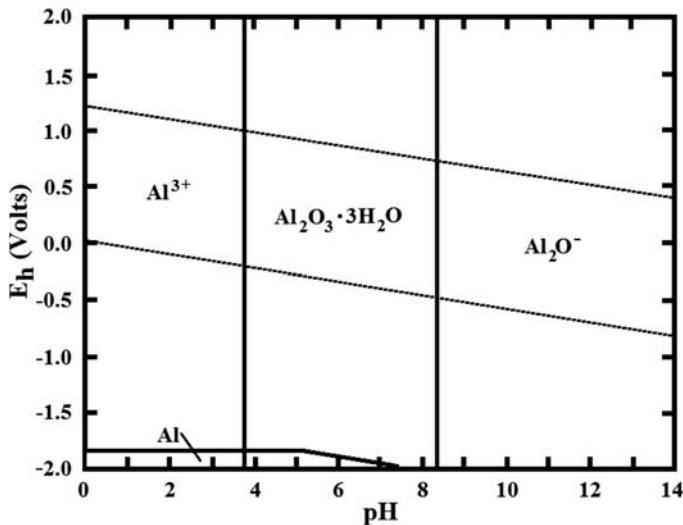


FIGURE 5.3-27 Pourbaix diagram of aluminum–water system. Adapted from M. Pourbaix, *Atlas of Electrochemical Equilibria in Aqueous Solutions*, National Association of Corrosion Engineers, Houston, Texas, 1974. Reproduced by permission of National Association of Corrosion Engineers.

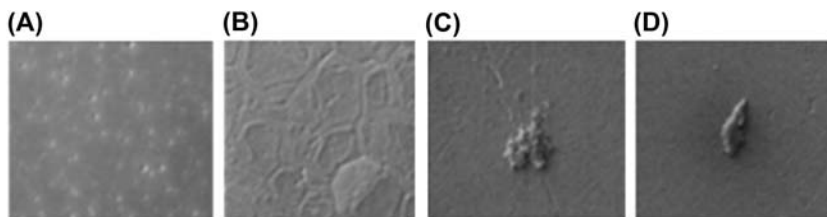


FIGURE 5.3-28 SEM images of (A) aluminum corrosion pits, (B) grain boundary decoration, (C) scratch-related alumina residue, and (D) aluminum organic compound/residue on aluminum surface. *Used with permission from the authors.*

particles. In acidic solutions, the zeta potentials of alumina particles are in the positive region. In terms of particle removal or particle reattachment reduction, it is preferable to use alkaline solutions. However, pitting or grain boundary decoration is observed on Al surface after cleaning with high-pH solutions. Fig. 5.3-28 shows examples of SEM images of defects such as corrosion pits, grain boundary decoration, and residues on the Al surface. Fig. 5.3-29 shows wafer defectivity maps denoting different defect signatures for scratches, residue, and circular brush marks. Dilute citric acid-based or oxalic acid-based ($C_2H_2O_4$) cleaning solutions have been found to be applicable to Al post-CMP cleaning, typically without serious corrosion issue. Additives can be added into acidic solutions to modify Al_2O_3 particle zeta potential to reduce brush particle loading.

5.3.5 Copper/Low-k Dielectric Post-CMP Cleaning Solutions

Copper post-CMP cleaning solutions typically contain organic acids, organic salts or a quaternary ammonium hydroxide (R_4N-OH , where R = organic moiety), chelating agents, surfactants, and organic solvents. As a result of the presence of various organic additives in Cu CMP slurries and polishing byproducts, organic residue deposits on the wafer surface and their removal is

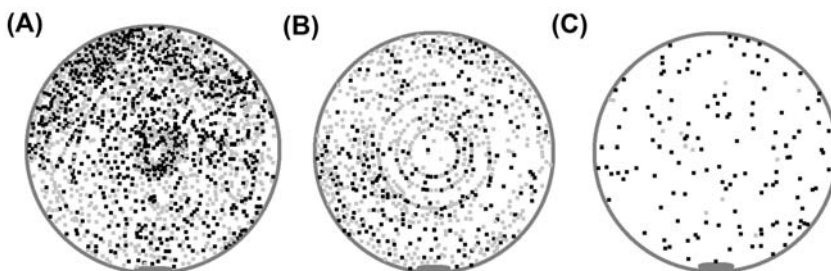


FIGURE 5.3-29 Examples of Al wafer defect scan maps with (A) polish scratches and residues, (B) brush circular scratches, and (C) a scan map with random defect distribution. *Used with permission from the authors.*

a challenge for cleaning [54]. Acidic solutions exhibit better cleaning performance than the early generations of alkaline solutions, in terms of organic residue removal and smooth Cu surface generation. Citric acid- or oxalic acid-based Cu post-CMP cleaning solutions are commonly used. For BEOL post-CMP cleaning, typically concentrations of commercial cleaning solutions ranging from 1 to 2 wt% are used, although the concentrations of active ingredients are significantly lower. Oxidizers such as H_2O_2 cannot be used for metal post-CMP cleaning solutions because of corrosion issues.

Particle adhesion force measurement with atomic force microscopy shows a low adhesion force in citric acid solution with the addition of NH_4OH [55]. Various additives are added into the acidic solution to improve particle removal performance. For advanced nodes, the implementation of new Cu barrier materials, where the metal stack sidewall is vulnerable to chemical attack and an increased reactivity of narrow Cu lines that are also vulnerable to corrosion, requires alkaline cleaning chemistry. TMAH-based and non-TMAH-based alkaline solutions are the dominant chemistry for this application [56,57]. An electrochemical impedance spectroscopy analysis [58] indicates that a minimum concentration of 0.5 wt% TMAH is required for BTA removal from Cu surface. Cu surfaces are better protected from further oxidation because of the presence of a cuprous oxide (Cu_2O) layer [59,60], as can be seen in the simplified Cu– H_2O system Pourbaix diagram on Fig. 5.3-30 [9].

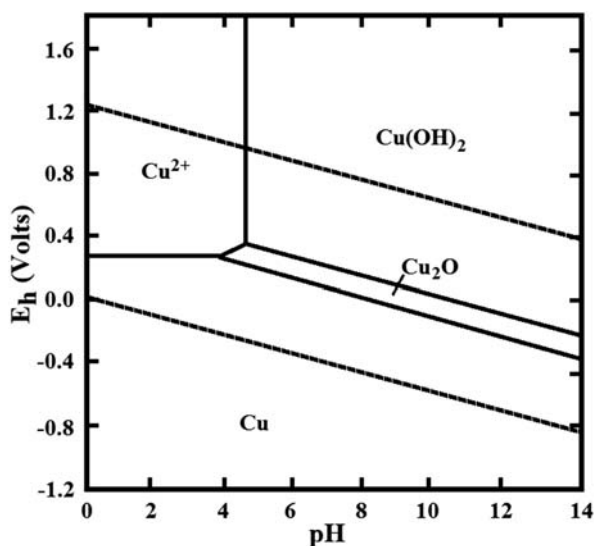


FIGURE 5.3-30 Potential–pH diagram of Cu–water system. Adapted from M. Pourbaix, *Atlas of Electrochemical Equilibria in Aqueous Solutions*, National Association of Corrosion Engineers, Houston, Texas, 1974; E.A. Kneer, C. Raghunath, V. Mathew, S. Raghavan, J.S. Jeon, *J. Electrochem. Soc.*, 144 (9) (1997) 3041. Reproduced by permission of National Association of Corrosion Engineers.

Furthermore, a more favorable film surface zeta potential is generated in a high-pH environment. However, the particle-attracting cuprous oxide surface after alkaline cleaning makes it more difficult to prevent particle reattachment. Post-CMP Cu surface roughness depends on slurries and cleaning solutions. Fig. 5.3-31 shows a schematic of Cu surface layer after cleaning with acidic and alkaline solutions. Fig. 5.3-32 shows zeta potential of Cu surface after cleaning. With new-generation alkaline cleaning solutions, post-CMP Cu surface roughness in the range of 0.4–0.6 nm typically is achieved. Chapter 10 discusses the chemistry and morphology of the Cu surface, including passivation.

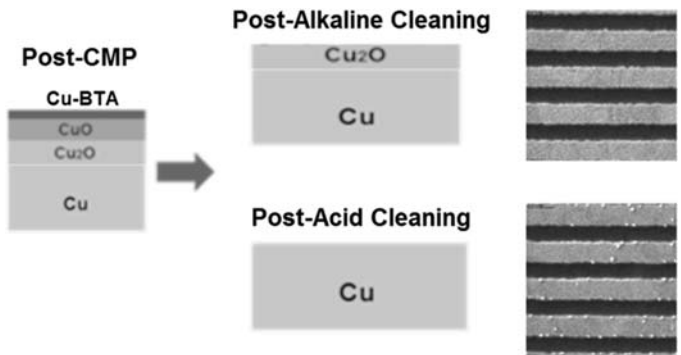


FIGURE 5.3-31 Schematic of Cu surface after cleaning with acidic and alkaline solutions. SEM images show the comparison of Cu surface after cleaning with an alkaline and acidic solution. Cu_xO particle growth appeared on unprotected Cu surface associated with acid cleaning. *Used with permission from the authors.*

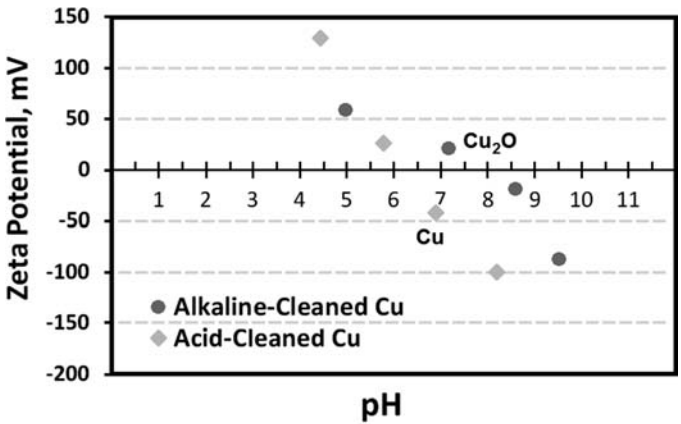


FIGURE 5.3-32 Zeta potential of Cu surface after cleaning with an acidic and alkaline solution. The IEP of Cu surface is at pH about 6 and 8, respectively, without or with the presence of Cu₂O layer on Cu surface. *Used with permission from the authors.*

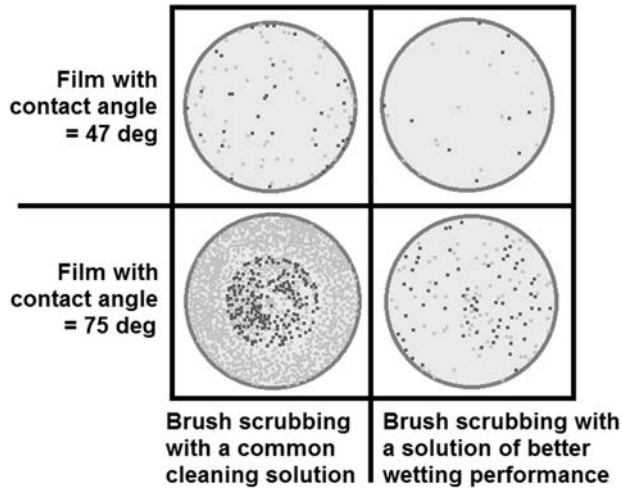


FIGURE 5.3-33 Example of particle scan maps of dielectric films with different water contact angles. For a dielectric film with DIW contact angle at 47 degrees, brush scrubbing with a common post-CMP cleaning solution did not cause significant particle addition onto film surface (upper left scan map). For a film with DIW contact angle of 75 degrees, scrubbing with the common solution generated a data-overloaded particle scan map, >60,000 particles (lower left scan map). Brush scrubbing the two films with a cleaning solution with better film surface wetting performance showed significant particle performance improvement (the two scan maps on the right side). *Used with permission from the authors.*

Low-k dielectric films are hydrophobic. To avoid k-value increase, the hydrophobic surface is not allowed to be converted to a hydrophilic surface. Poor surface wetting by cleaning solutions may cause high particle adds, as shown with the wafer maps in Fig. 5.3-33. During brush scrubbing, residues are transferred from the brushes to the wafer surface. Particle attachment may also occur during wafer transfer from the cleaning chamber to the drying chamber. To improve low-k film surface wetting, surfactants are commonly added into cleaning solutions.

Depending on the CMP slurry, the contact angles of water on polished low-k film may vary from 45 to 80 degrees. Particle tests with low-k dielectric films of different contact angles indicate that for films with low hydrophobicity (contact angles approximately 50 degrees or lower), common Cu post-CMP cleaning solutions can be used. But for wafers with high surface hydrophobicity (contact angle about 80 degrees), particles on the brush would be pressed onto the hydrophobic surface, indicated in Fig. 5.1-11 by the particle scan data overloaded on the scan map. With a cleaning solution containing a surface wetting agent, brush scrubbing can be used to achieve slurry particle removal. For post-CMP cleaning of hydrophobic films, the addition of appropriate surfactants in the cleaning solutions is critical [61,62].

Post-CMP cleaning of low-k dielectric films is considered a significant challenge. As a result of the implementation of Marangoni IPA drying for post-CMP cleaners, the challenge of watermark generation is overcome. However, because of the continuous reduction of the dielectric constants of low-k films and with their increasing porosity and carbon content, there is a renewed concern about tool and process capability for ultralow-k materials (k values at 2.5 or below) post-CMP cleaning, in particular prevention of low-k film damage. Particles on the brushes or in the liquid can adhere onto the low-k surface, requiring nanometer-size particle removal.

The contact angle of a DIW droplet on an as-deposited hydrophobic low-k film is typically greater than 80 degrees. The addition of surfactants into post-CMP cleaning solutions improves wetting performance on hydrophobic surface. However, water and surfactant can penetrate into the nano-pores of the low-k material. Therefore, surfactants must be removed after cleaning; otherwise, the uptake or absorption of surfactants will cause a k-value increase [63–65]. Fig. 5.3-34 shows a comparison of Fourier transform infrared spectra of an organosilane low-k dielectric film before and after cleaning with a cleaning solution.

In addition to particle and organic residue removal capability, metal contaminant removal capability, as determined by TDDb performance, is a key factor used for selecting cleaning solutions for Cu post-CMP [65]. With the addition of proper chelating agents, typically, metallic contaminants below

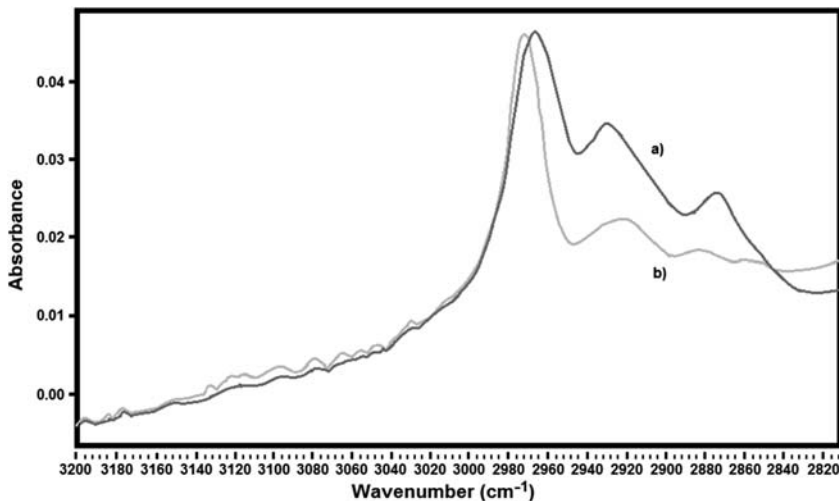


FIGURE 5.3-34 Fourier transform infrared spectra of an organosilane low-k dielectric film (a) after cleaning with a Cu post-CMP cleaning solution causing k-value increase compared with (b) an unprocessed, as-deposited low-k film. *Used with permission from the authors.*

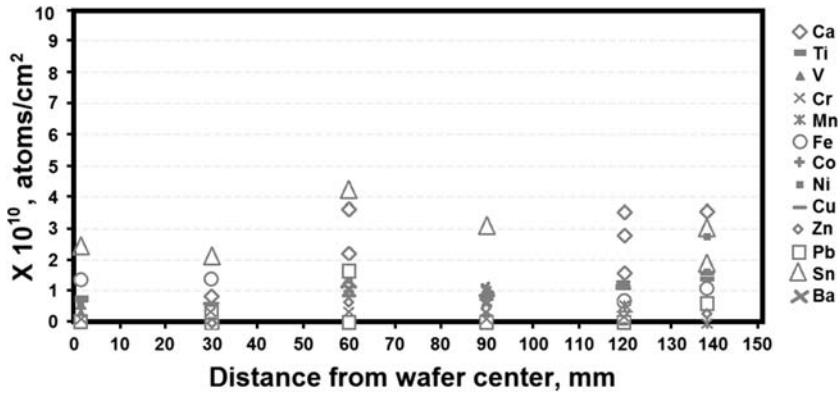


FIGURE 5.3-35 Example of TXRF analysis of a SiO₂ film on silicon wafer after Cu post-CMP cleaning. Typically, metal contaminant concentrations below 5×10^{10} atoms/cm² across wafer surface can be achieved after Cu post-CMP cleaning. *Used with permission from the authors.*

5×10^{10} atoms/cm², analyzed using TXRF or VPD ICP–MS, can be achieved with either acidic or alkaline cleaning solutions. Fig. 5.3-35 shows an example of TXRF analysis of SiO₂ film on a silicon wafer after Cu post-CMP cleaning. With strict post-CMP queue time control, TDDb performance of Cu post-CMP cleaning with alkaline and acidic chemistries are comparable; however, cleaning with alkaline solutions provides a much broader queue time window before capping layer deposition.

5.4 POST-CMP CLEANING EQUIPMENT

Brush scrubbing [66] is the dominant cleaning method used for post-CMP cleaning. Typically, two brush boxes are used; the first brush box is dedicated to heavy-duty cleaning and the second brush box is dedicated to fine cleaning. A single cleaning solution or two different cleaning solutions can be used in the two brush boxes. The front surface and the back surface of the wafer are scrubbed with two PVA roller brushes, one on each side. Brush compression force, brush and wafer rotation speed, and liquid flow rates through the brush core and the brush box spray bars are the main process parameters [67]. In general, higher compression and thus higher friction force of the brush on the wafer surface provide higher particle removal efficiency. However, if brushes are loaded with particles, higher brush compression may cause more particle reattachment onto the wafer surface. Therefore, optimal brush compression is a compromise between particle removal efficiency improvement and particle reattachment reduction.

Megasonic cleaning is also widely used for post-CMP cleaning [16]. Typically, a frequency of 950 kHz is used. As a noncontact cleaning method,

megasonic cleaning shows good capability for removing residues from the recessed area where brush scrubbing cannot reach. Megasonic cleaning at elevated temperature (40 to 50°C), compared with room temperatures, shows better capability for removing particles and organic residues, which are difficult to remove by brush scrubbing. Megasonic cleaning with diluted SC-1 solution is a common cleaning process in FEOL CMP. For BEOL CMP, megasonic cleaning is focused on organic residue removal.

A PVA sponge or pad in the form of a disk can be used for post-CMP cleaning [68], either before or after a brush box. In comparison with roller brushes, the concern about the PVA disk is its shorter cleaning time across the wafer surface and potentially shorter life of the PVA disk, compared with roller brushes. Because of the similar cleaning mechanism, both PVA pad scrubbing and PVA roller brush scrubbing have similar cleaning capability for particles and organic residue removal from the wafer's front surface. PVA disks show promise in addressing the limitation of roller brushes and also in supplementing the capability of roller brush cleaning.

Jet spray has received limited use for post-CMP cleaning [16], although there are potential benefits using jet spray as a noncontact cleaning method. In comparison with brush scrubbing, jet spray cleaning typically shows relatively low efficiency for removing strongly adhered CMP slurry particles and organic residues. However, high-velocity jet spray may help to remove particles in recessed area where brush scrubbing cannot reach, but strong jet spray may not be applicable to Cu and hydrophobic low-k dielectric film cleaning. Less aggressive jet spray conditions provide an effective rinsing that is useful in removing loosely adhered particles that are deposited during the brush scrubbing step.

Single-wafer SRDs in vertical or horizontal configurations were originally used for post-CMP cleaning [16]. During the wafer spinning process, thick water layers are removed, layer-by-layer, from the wafer surface. Increasing wafer rotation speed generates higher centrifugal force and quickly spins off the water plus the residues dissolved or suspended in the water. However, the higher rotation speed may generate turbulence inside the SRD chamber and an increased number of particle adds onto the wafer surface. The final thin layer of water is removed from wafer surface through evaporation; as a result of this phenomenon, small particles inside the water layers may be left behind and remain on wafer surface. Ultraviolet lamp heating may be used to accelerate water evaporation but cannot be applied to dry patterned Cu wafers because of concern about the occurrence of Cu photo-induced corrosion. With the implementation of hydrophobic low-k dielectric film that is subject to water-mark generation, the application of SRD could not be extended to these films. The single-wafer-based, nonspin, Marangoni IPA dryers or IPA-assisted spin-based Rotagani dryers have replaced the SRD in most applications from the 90-nm technology node onward.

5.4.1 Brush Scrubbing Cleaning Modules

There are two types of brush scrubber configurations: horizontal and vertical, shown in Fig. 5.4-36. Each configuration has its pros and cons. The horizontal configuration retains liquid longer on the wafer surface, but residues in the brush can drip onto the wafer surface after cleaning. The vertical configuration requires uniform liquid dispense to achieve good surface coverage, but after scrubbing, once the two brushes are open and move away from the wafer surfaces, because no brush is above the wafers, residues and water on the brushes cannot drip onto the wafer. Additionally, with a vertical brush configuration, the cleaning solutions can be better retained onto the wafer back surface compared with the bottom horizontal brush, and thus, better cleaning efficiency can be achieved. Through optimization of the brush box hardware and the scrubbing process, similar cleaning performance can be achieved with both horizontal and vertical brush boxes. Delivering chemical through the brush core is an option [69], which may lead to more uniform chemical distribution from wafer center to edge. However, chemicals flowing through the brush core may leach out particles inside the brush and result in a large numbers of particle adders on wafer surface. This is observed with newly installed brushes, thus causing particle qualification failure of the CMP tool. Although the water flowing through the brush core and the chemical dispensed from the spray bars may leach out particles from brushes, particle release is slower with this configuration. When dispensing chemical through the brush core, wafer rinse capability using the same brush is compromised as a result of the brushes being soaked in the cleaning chemicals.

Particle removal efficiency of double-sided brush scrubbing depends on scrubbing friction force, scrubbing hydrodynamics, and chemical concentration. Friction force is controlled either by the selection of brush compression and overlap distance or by direct application of compression force [70]. Friction is also a function of wafer and brush rotation speed and liquid flow rates.

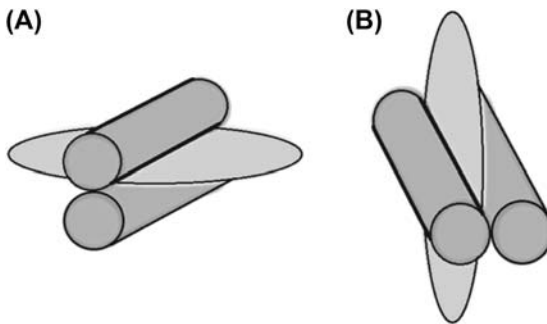


FIGURE 5.4-36 Schematic of horizontal (A) and vertical (B) double-sided brush scrubber. *Used with permission from the authors.*

The friction force and kinetics during brush scrubbing widely have been applied to explain phenomena observed in post-CMP cleaning. Brush scrubbing hydrodynamic and tribological studies [71–79] provide an understanding of particle removal mechanism, in terms of mechanical actions. Particle adders from cleaning tests may result from the combined phenomena of particle removal and particle reattachment [80]. When brushes are heavily loaded with contaminants, particle reattachment can cause more particles to be added than particles to be removed. Fig. 5.4-37 shows wafer particle scan maps with particle reattachment. Brush scrubbing process optimization requires both mechanical and chemical factors to be taken into consideration to maximize particle removal efficiency (without generating scratches on soft Cu and Al films) and minimize particle reattachment in a high-volume manufacturing environment.

The control of wafer and particle zeta potential plays an important role in particle removal with brush scrubbing [81]. The particle scan maps after brush scrubbing with DIW, shown in Fig. 5.4-38, indicate that particles can be removed

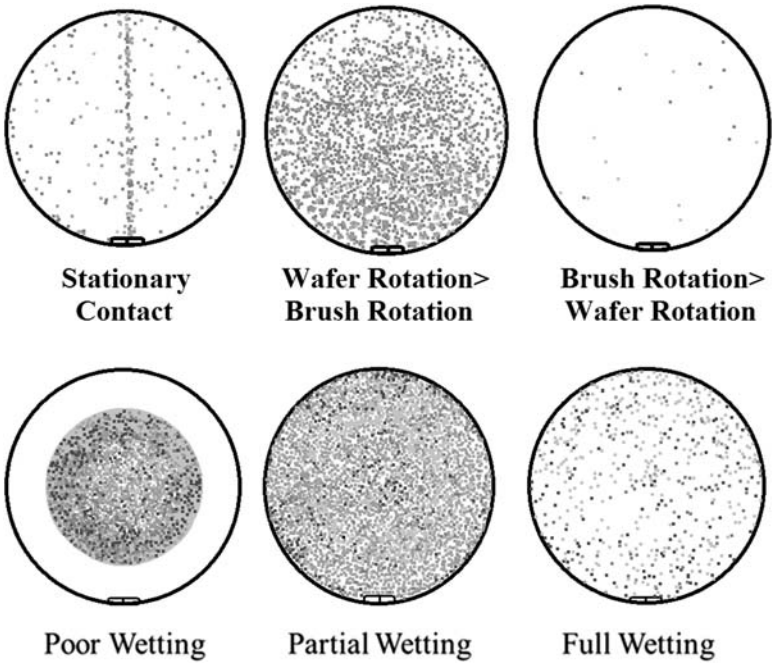


FIGURE 5.4-37 Effect of brush–wafer rotation velocity and wafer surface property on particle transfer from brushes onto wafer surface during brush scrubbing. Depending on relative motion between wafer and brushes, particles on brushes could be transferred back onto wafer surface (brush marks showed up in upper two scan maps). Particle reattachment could be more serious for brush scrubbing with liquids of poor film surface wettability (particle overloaded in the lower left particle scan map). *Used with permission from the authors.*

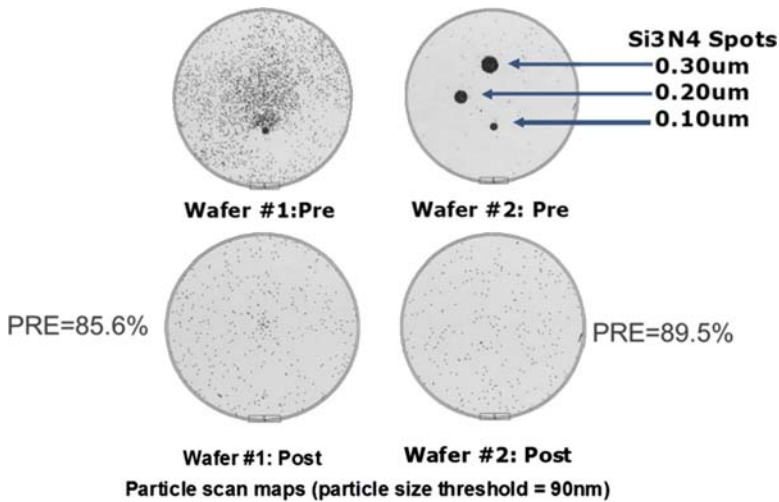


FIGURE 5.4-38 Silicon nitride particle removal from silicon dioxide film on silicon wafers by brush scrubbing with DIW. Brush scrubbing with DIW, which is mechanical action dominated, is able to remove particles. However, many particles are relocated or reattached on wafer surface, because of electrostatic attraction between silicon nitride particles and silicon oxide film surface. *Used with permission from the authors.*

or relocated by scrubbing even without using chemicals. However, because of the positive zeta potential of Si_3N_4 particles, the positively charged particles may stay on the brush surface and reattach back onto the wafer. In FEOL post-CMP cleaning, low concentrations of alkaline chemistries, such as 1:1000 dilution of 28 wt% NH_3 aqueous solution with H_2O , are typically sufficient for silica slurry particle removal, implying that the main role of the alkaline chemistries is to create the required zeta potential and electrostatic repulsive forces, rather than to generate particle undercutting and particle lift-off.

5.4.2 Megasonic Cleaning Modules

Although megasonic nozzles, rods, or bars [82,83] have specific applications with respect to particle removal, it is the single-wafer, immersion tank megasonic cleaning module that has gained widespread application for post-CMP cleaning in high-volume manufacturing. In the single-wafer immersion megasonic module, the megasonic transducer is placed at the bottom of a tank and the wafer is rotated by rollers in which the wafer is also held in place. As a result, the wafer's front surface, back surface, and bevel edge are cleaned. Fig. 5.4-39 shows a schematic of a single-wafer immersion megasonic module. The typical megasonic frequency is 950 kHz. Through the control of megasonic power (or power density), megasonic energy is supplied from the tank bottom. Liquid flows from tank bottom to top and flows out through the overflow weir.

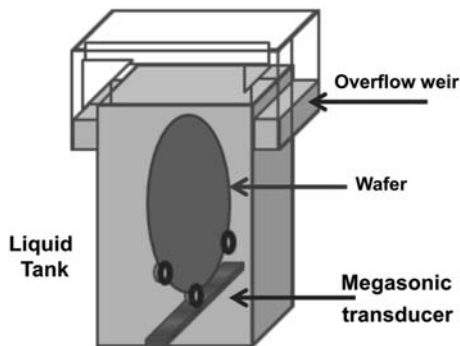


FIGURE 5.4-39 Schematic of a single-wafer immersion tank megasonic module. *Used with permission from the authors.*

With an immersion tank, megasonic energy loss can be minimized, as a result of the energy transfer through the liquid inside the tank, because there is always liquid between the megasonic transducer and the wafer, that is, there is no air exposure. Because megasonic particle removal efficiency depends on the chemical concentration, temperature, and process time for particle undercutting, plus the power density and frequency for achieving acoustic streaming, these are the main process parameters. Megasonic cleaning before brush box cleaning helps to reduce particle loading onto the brushes. This technology is extendable to advanced node FEOL and BEOL post-CMP cleaning. For BEOL Cu CMP, to avoid generating Cu surface roughness, megasonic cleaning with reduced chemical concentrations or with DIW alone is used.

5.4.3 PVA Disk Cleaning Module

For the PVA disk cleaning module, shown in [Fig. 5.4-40](#), either a vacuum chuck or edge grippers are used to hold the wafer in the vertical position. While the wafer is rotated, the rotating PVA disk is pressed onto the wafer and swept across its surface [\[81\]](#). A cleaning solution or DIW is dispensed through a spray nozzle onto the wafer. The implementation of wafer back surface or edge cleaning is difficult using a PVA disk module in this configuration. However, wafer front surface scrubbing with a small PVA disk provides a capability similar to that achieved with roller brushes. For rinsing, the centrifugal force is an advantage, allowing high wafer and/or PVA disk rotation speed to be used. Also, because the contact between PVA disk and wafer surface is flat, this method generates lower risk of scratching wafers than roller brush scrubbing. The process parameters for PVA disk brush cleaning include wafer and disk rotation speed, downforce, sweep speed, and chemical

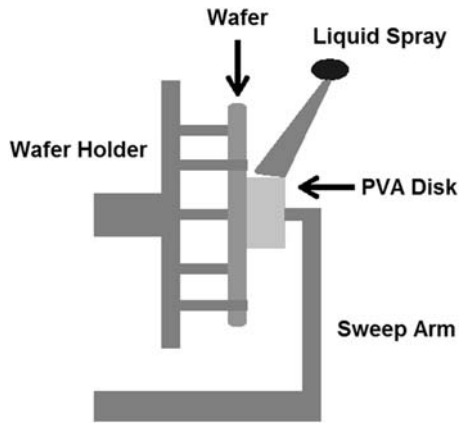


FIGURE 5.4-40 Schematic of a PVA disk brush scrubbing module. *Used with permission from the authors.*

concentration. Cleaning wafers with a PVA disk before the brush boxes helps to reduce circular scratches generated by roller brushes. For fine cleaning, PVA disk scrubbing after brush box cleaning may help to mitigate particle issue associated with roller brushes, such as brush particle loading or residue dripping from the roller brush to the wafer.

5.4.4 Jet Spray Cleaning Module

Jet spray cleaning, shown in Fig. 5.4-41, is a noncontact cleaning approach [84,85]. This method has attracted interest for use in post-CMP cleaning, especially when a comparison is made with roller brush cleaning that requires a large numbers of dummy wafers for brush break-in and that has a high

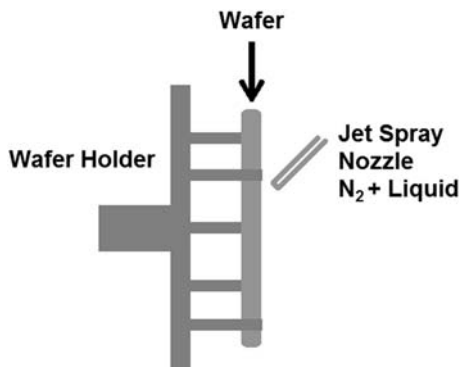


FIGURE 5.4-41 Schematic of a jet spray cleaning module. *Used with permission from the authors.*

incident of particle reattachment on the wafer surface during brush contact. However, particle or organic residue removal efficiency with jet spray cleaning is typically lower than brush scrubbing. Jet spray is created by mixing a cleaning solution or DIW with a carrier gas, such as N_2 . Carbon dioxide may be added into DIW to increase liquid conductivity. Jet spray and wafer surface interactions may generate a splashing effect [86,87] that results in localized de-wetting of hydrophobic films and thus may generate watermarks [16]. Jet spray cleaning has not been proven to be an alternative post-CMP cleaning method for replacing brush scrubbing. Chapter 4 outlines the various jet spray methods.

Jet spray particle removal efficiency depends on the spray velocity, the droplet size, the distance between spray nozzle and wafer surface, and the concentration of the cleaning solution. The N_2 carrier gas flow rate and liquid flow rate, which control the spray velocity and the droplet size, are key process parameters to control. Although jet spray is not as effective as brush scrubbing for particle and organic residue removal, jet spray can provide better wafer rinse performance than spin rinse.

5.4.5 Drying Modules

SRDs use centrifugal force to remove water from the wafer surface. For hydrophilic surfaces, high centrifugal forces generated by high wafer rotation speed are helpful for water removal. For hydrophobic surfaces, low wafer rotation speed is helpful in reducing water beading up. However, because centrifugal force varies along wafer radius direction, i.e., the center of the wafer sees less centrifugal force than the edge, it is difficult to prevent water from beading up or splashing. Also, too low rotation speed may not generate enough centrifugal force at the wafer edge to spin away residue-containing water from the wafer surface. For patterned device wafers, because of the coexistence of hydrophilic and hydrophobic surfaces for structures that include different pitch sizes and pattern densities, watermark generation caused by SRD drying cannot be eliminated.

Marangoni drying [88] with IPA is based on the surface tension gradient at the wafer surface in contact with the water meniscus. Because of the low surface tension of IPA, 27 dyn/cm, versus H_2O , 72.8 dyn/cm at 20°C, the surface tension of the bulk water is higher than that of the water at the meniscus because of the dissolved IPA. The force of the surface tension gradient resulting from the IPA concentration gradient drags the water layer on the wafer surface into the bulk water of the bath, when a wafer is pulled out of the liquid in a vertically configured Marangoni dryer. The Marangoni dryer does not rely on the displacement of water by IPA, such as an IPA vapor dryer does. Marangoni effect only depends on the surface tension gradient in the water and does not rely on the absolute values of the surface tension or IPA concentration of the IPA–water mixture. Therefore, the IPA usage is low, for

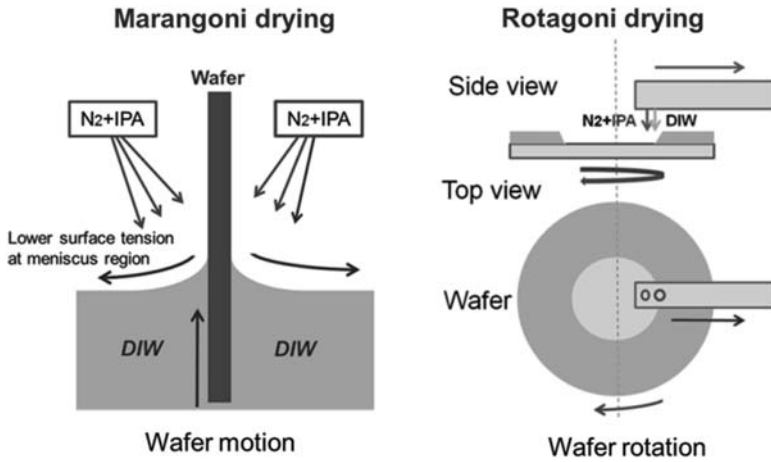


FIGURE 5.4-42 Schematic of a Marangoni drying module and Rotagoni drying module. Used with permission from the authors.

example less than 1 g of IPA liquid per 300-mm wafer has been obtained. Because there is no wafer spinning, the drying occurs in the horizontal zone of wafer–water interface. Fig. 5.4-42 shows a schematic of a Marangoni dryer.

Constant IPA concentration in the N_2 carrier gas can be supplied to the wafer–water meniscus by the injection of IPA liquid into a vaporizer. In the single-wafer configuration, high lift speed of the water out of the bath and the minimized contact between the wafer edge and the wafer carrier make it possible to achieve high throughput. IPA Marangoni drying provides effective drying of hydrophobic and hydrophilic surface and eliminates watermarks. Also, unlike SRD, Marangoni drying does not generate surface charging.

A horizontal configured Rotagoni [89] dryer is also shown in Fig. 5.4-42. While the wafer is rotated and the dry–wet interface is moving outward after starting at the center, the IPA vapor in the N_2 carrier gas is dispensed at the interface to assist wafer drying and to reduce watermark generation. For highly hydrophobic films, it is difficult to control the moving interface during wafer rotation. Water splashing back onto the dried area causes particle issue. However, low-k film surface hydrophobicity is reduced after polishing, presumably because of the adsorption of surfactant from the CMP slurries. Rotagoni drying has been applied in drying the low-k films with reduced hydrophobicity in post-CMP cleaning.

5.5 FUTURE DIRECTION OF CMP AND POST-CMP CLEANING

As more CMP steps are being added into the fabrication of advanced devices and because of the shrinking geometries and new materials being introduced,

the CMP and post-CMP processes are required to meet more stringent requirements with respect to planarity, uniformity, defectivity, and productivity and at reduced cost [90]. For 3D NAND memory, high removal rate with high planarization efficiency is required for thicker SiO₂ film removal for the staircase structures [91]. The slurries for these devices contain more abrasive particles compared with previous generations of slurries [92], leading to post-CMP cleaning that requires a significant improvement in particle removal efficiency. For thinner film removal, to match the short polishing time, post-CMP cleaning must achieve high throughput, which typically is limited by the dryers. High removal rate selectivity drives the need for achieving sub-nanometer range within-wafer and within-die nonuniformity. The use of advanced slurries with additives that are used to suppress or accelerate the removal rate of specific films and to modify the film surface property both contribute to the organic residue that remains after polishing. New materials used at advanced nodes require CMP slurries and post-CMP cleaning solutions to have better materials compatibility, including reduction in the film etching rates and prevention of galvanic coupling between the barrier, liner, and metal films.

Post-CMP cleaning process development is typically based on wafer-level defect data from statistical design of experiments, which help understand the effect of the cleaning process parameters on the results, such as particle removal efficiency and defectivity. Insight of the chemical reactions involved in polishing and the understanding of the reaction products help to produce a comprehensive understanding of the physical and chemical phenomena of the slurry-cleaning chemical interaction. For a specific slurry, there may be a corresponding post-CMP cleaning solution that provides optimum cleaning performance. The traditional strategy of using one cleaning solution or one best-known cleaning process for all types of slurry may not provide an optimum cleaning process that meets the CMP defect requirements at small particle sizes. The removal of abrasive particles and organic residues requires integration of polishing and cleaning sequences/processes. On the one hand, polishing process optimization for reducing residue carry-over to post-CMP cleaner is needed. On the other hand, development of post-CMP cleaning solutions is required to take into consideration the chemistry of the specific CMP slurry and polishing byproducts.

5.5.1 New Post-CMP Cleaning Technology

To achieve particle reduction, post-CMP cleaning focuses on particle removal efficiency improvement and particle reattachment prevention. As the dominant post-CMP cleaning method, brush scrubbing is challenged as a result of its low efficiency in removing strongly adhered particles and the high chance of causing particle reattachment during brush contact with wafer. Additional cleaning after brush scrubbing has demonstrated reduced particle reattachment. Post-CMP

cleaning must achieve high particle removal efficiency with low chemical and mechanical aggressiveness, eliminate corrosion, minimize surface roughening, and reduce scratching. Although new post-CMP cleaning capability is demanded for meeting defect reduction requirements, a question remains with respect to what available cleaning technologies will be suitable to address the cleaning challenge for advanced nodes. CMP is based on the chemical and mechanical action at the contact area of film, chemicals, slurry particles, and polishing pad to achieve film material removal. For Ce particle removal, a wafer buff process, that is, polishing a wafer with DIW on a polisher platen, which is a purely mechanical cleaning process, is commonly applied to reduce particle counts.

To make use of the uniform, nanoscale contact between film, pad, and chemicals, a CMP-type cleaning method, chemical mechanical cleaning (CMC) [14], is able to achieve a significant improvement in particle removal efficiency. This cleaning technology is implemented in a chemical buff [79] or CMC module [14]. The wafers are “polished” with a CMP pad and cleaning solution, which can remove strongly adhered particles before wafers are scrubbed in the brush boxes [10]. As it is shown in Fig. 5.5-43, brush scrubbing and megasonic cleaning cannot remove the organic residues generated by polishing with a challenging Cu barrier slurry, CMC provides the capability of removing the residues from Cu surface. As Ce particles receive more applications for CMP, the implementation of CMC for post-CMP cleaning significantly raises particle removal efficiency for integrated post-CMP cleaners [14]. Fig. 5.5-44 shows the application of CMC for ceria slurry particle removal.

5.5.2 Post-CMP Cleaning of New Materials

Cobalt Post-CMP Cleaning: As device geometry further shrinks to the 7- and 5-nm nodes, Co outperforms W, when used for local contacts, and Cu, when used as an interconnect material, with respect to the low resistance that is

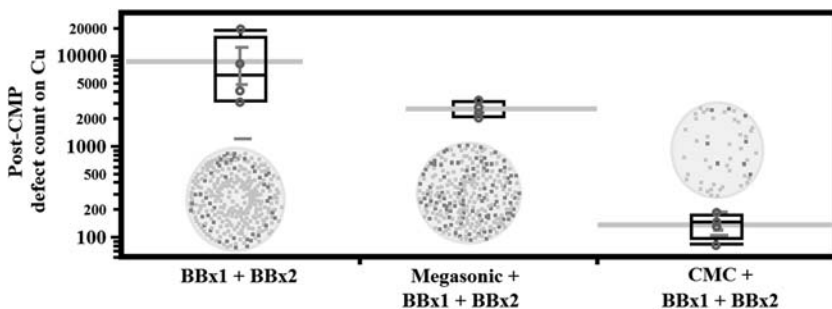


FIGURE 5.5-43 Chemical mechanical cleaning (CMC) for slurry residue removal from Cu polished with a challenging Cu barrier slurry containing high content of organic additives. Wafer defect scan maps showed around 8000 defects after brush scrubbing. Adding Megasonic cleaning showed limited improvement in particle/residue removal. Adding CMC significantly reduced defects to less than 100 counts. *Used with permission from the authors.*

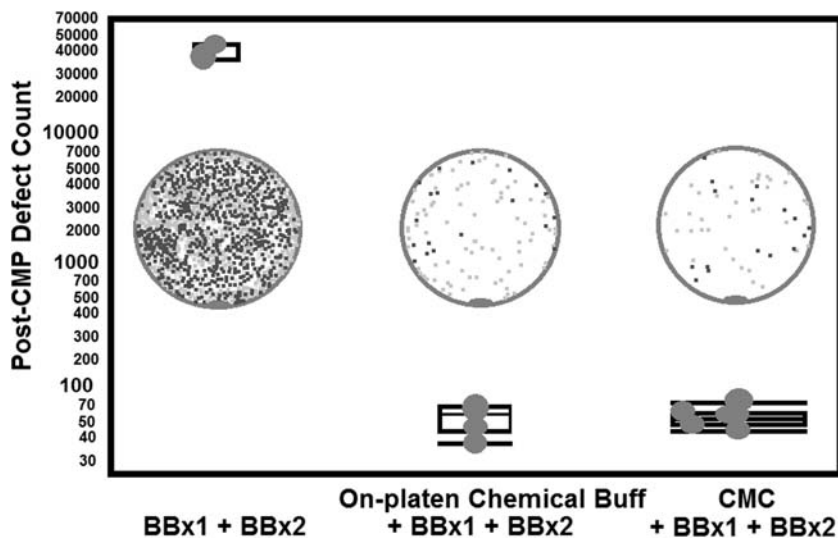


FIGURE 5.5-44 Chemical mechanical cleaning (CMC) for ceria slurry particle removal. Wafer particle scan maps after brush scrubbing showed 40,000–60,000 particle counts, indicating low efficiency for ceria particle removal. Adding on-platen chemical buff or CMC in post-CMP cleaner significantly reduced post-CMP defect counts. *Used with permission from the authors.*

required for the reduced geometry and film thickness [93–95]. Therefore, Co CMP [96] is being expanded from Cu barrier CMP, where Co is used as the seed enhancement layer in the sub-10-nm node, to where Co is used as the contact material and where Co is used as an interconnect material in 10-nm node or below. Because of the high reactivity of Co and the smaller geometry of Co lines or plugs, corrosion prevention in Co CMP and post-CMP cleaning is a challenge [97].

Silicon germanium post-CMP cleaning: High-mobility materials are being considered for replacing Si in the transistor channel: III-V compound semiconductor materials [98] with high electron mobility for n-type channel field effect transistors and silicon germanium (SiGe) with high hole mobility for p-type channel field effect transistors. It is expected that CMP can be an enabling technology for the integration of these new materials [99–101]. Post-CMP cleaning of SiGe will be critical for the implementation of SiGe channel. Without an oxidizer, diluted NH_4OH does not significantly etch SiGe and may be considered for particle removal. Also, when the oxidizer is absent, diluted HF does not generate significant SiGe loss, but can etch silicon oxides and some germanium oxide species [102,103]. HF may still be needed to remove metallic contamination during post-CMP cleaning. Cleaning performance on the wafer level must be studied for different CMP slurries. It is expected that improving particle removal efficiency and minimizing film material loss will be the focus in post-CMP cleaning for SiGe structures.

5.6 SUMMARY

Since approximately 1996, CMP has become an enabling technology in the implementation of new materials for new device structures with shrinking geometry. In CMP, selective material removal and surface topography planarization are achieved through the use of slurries with different abrasive particles and chemical additives. For post-CMP cleaning, slurries act as contaminants. Further, during polishing, the slurry contaminants and polishing byproducts are pressed onto wafer surface, which makes contaminant removal more difficult. This is a unique challenge of post-CMP cleaning. Accordingly, post-CMP cleaning has evolved into a unique segment of semiconductor cleaning and CMP-specific cleaning technologies.

CMP applications include polishing of various materials— SiO_2 , Si_3N_4 , polySi, and low-k dielectric films; to W, Al, Cu, and other metals; and various diffusion barrier and functional materials, such as TiN. Different chemical and mechanical phenomena are involved in the polishing of different film materials. Different polishing byproducts and residues are generated on film surfaces with different properties (e.g., hydrophilic, hydrophobic, soft, hard, passivated, and chemically active). The complex mechanical, chemical, and electrochemical phenomena occurring on the wafer surfaces during wafer polishing, wafer transferring, and post-CMP cleaning in a dynamic environment require much more fundamental studies to quantify the specific conditions of CMP defect generation. One example is Cu dendrite growth on specific pattern structures under certain conditions. CMP defect reduction and post-CMP cleaning are now progressing from an art to a scientific and engineering segment with established fundamental understandings and engineering quantification methods.

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